# Test du package levyfit

## Sommaire

**Préparation** **1**

**I Une trajectoire** **4**
I.1 Cas 1 : inspections régulières . . . . . 4
I.2 Cas 2 : inspections croissantes . . . . . 8
I.3 Cas 3 : inspections décroissantes . . . . . 13
I.4 Cas 4 : inspections mixtes . . . . . 18

**II Plusieurs trajectoires** **23**
II.1 Cas 5 : inspections régulières . . . . . 23
II.2 Cas 6 : inspections croissantes . . . . . 27
II.3 Cas 7 : inspections décroissantes . . . . . 31
II.4 Cas 8 : inspections mixtes . . . . . 35

## Préparation

Installation du package depuis GitHub (à faire une seule fois, puis redémarrer R) :

```
remotes::install_github("rytracetito/levy_packages", subdir = "levyfit")
```

Nombre d'inspections et dernier instant d'inspection des cas 1 à 4 :

```
n <- 100
tmax <- 100
```

Vraies valeurs des paramètres :

```
vrai_gamma   <- c(a = 1.5, b = 3)
vrai_wiener  <- c(mu = 0.5, sigma2 = 2.25, s2 = 2.25)
vrai_ig      <- c(mu = 0.5, eta = 2, eta_unbiased = 2)
vrai_vg      <- c(theta = 0.5, sigma = 1.5, nu = 2)
vrai_meixner <- c(a = 1, b = 0.3, d = 1.5, m = 0.2)
vrai_igam    <- c(theta = 1, alpha = 1, beta = 1.5)
```

Simulation d'une trajectoire de chaque modèle aux instants `t`, avec les vraies valeurs ci-dessus :

```
sim_gamma <- function(t) {
  a <- vrai_gamma[["a"]]; b <- vrai_gamma[["b"]]
  c(0, cumsum(rgamma(length(t) - 1, shape = a * diff(t), rate = b)))
}

sim_wiener <- function(t) {
  mu <- vrai_wiener[["mu"]]; sigma <- sqrt(vrai_wiener[["sigma2"]])
  c(0, cumsum(mu * diff(t) + sigma * sqrt(diff(t)) * rnorm(length(t) - 1)))
}
```

```r
sim_ig <- function(t) {
  moy <- vrai_ig[["mu"]] * diff(t)
  forme <- vrai_ig[["eta"]] * diff(t)^2
  nu <- rnorm(length(moy))^2
  y <- moy + moy^2 * nu / (2 * forme) -
    moy / (2 * forme) * sqrt(4 * moy * forme * nu + moy^2 * nu^2)
  c(0, cumsum(ifelse(runif(length(moy)) <= moy / (moy + y), y, moy^2 / y)))
}

sim_vg <- function(t) {
  theta <- vrai_vg[["theta"]]; sigma <- vrai_vg[["sigma"]]; nu <- vrai_vg[["nu"]]
  g <- rgamma(length(t) - 1, shape = diff(t) / nu, scale = nu)
  c(0, cumsum(theta * g + sigma * sqrt(g) * rnorm(length(t) - 1)))
}

sim_meixner <- function(t) {
  p <- vrai_meixner
  c(0, cumsum(vapply(diff(t), function(s) {
    loi <- Runuran::udmeixner(alpha = p[["a"]], beta = p[["b"]],
                              delta = p[["d"]] * s, mu = p[["m"]] * s)
    Runuran::ur(Runuran::pinvd.new(loi), 1)
  }, 0)))
}

sim_igam <- function(t) {
  theta <- vrai_igam[["theta"]]; alpha <- vrai_igam[["alpha"]]; beta <- vrai_igam[["beta"]]
  du <- 1e-3
  G <- cumsum(rgamma(ceiling(1.5 * max(t) / theta / du), shape = du))
  while (G[length(G)] <= max(t) / theta) {
    G <- c(G, G[length(G)] + cumsum(rgamma(1e4, shape = du)))
  }
  c(0, alpha * (du * findInterval(t[-1] / theta, G))^(1 / beta))
}
```

Nombre de trajectoires des cas 5 à 8, chacune inspectée 10 fois, et simulation de ces trajectoires sous forme de liste de `data.frame(t, X)` :

```r
n_traj <- 20
simuler <- function(sim, t) lapply(1:n_traj, function(i) data.frame(t = t, X = sim(t)))
```

Représentation d'un plan d'inspection, d'une trajectoire, et tableau des estimations :

```r
tracer_plan <- function(t, titre) {
  ggplot(data.frame(t = t), aes(x = t, y = 0)) +
    geom_point() +
    labs(title = titre, x = "Temps", y = NULL) +
    theme_minimal() +
    theme(axis.text.y = element_blank(),
          panel.grid.major.y = element_blank(),
          panel.grid.minor.y = element_blank())
}

tracer <- function(t, x, titre) {
  ggplot(data.frame(t = t, x = x), aes(x = t, y = x)) +
    geom_line(colour = "grey60") +
```

```
    geom_point(size = 1) +
    labs(title = titre, x = "Temps", y = "Dégradation") +
    theme_minimal()
}

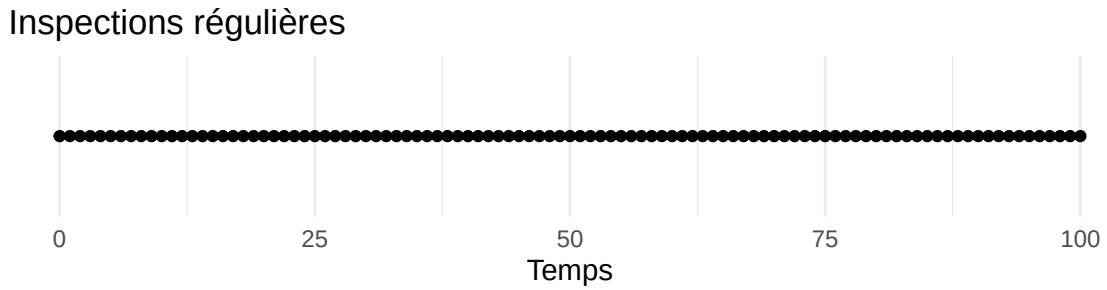
tracer_plusieurs <- function(liste, titre) {
  donnees <- do.call(rbind, lapply(seq_along(liste), function(i)
    data.frame(trajectoire = i, t = liste[[i]]$t, x = liste[[i]]$X)))
  ggplot(donnees, aes(x = t, y = x, group = trajectoire)) +
    geom_line(colour = "grey60") +
    geom_point(size = 0.6) +
    labs(title = titre, x = "Temps", y = "Dégradation") +
    theme_minimal()
}

resultat <- function(estimation, vraies) {
  if (inherits(estimation, "try-error")) {
    message <- conditionMessage(attr(estimation, "condition"))
    knitr::asis_output(paste("**Estimation impossible** :", message))
  } else {
    tableau <- data.frame(Paramètre = names(vraies), Vraie = vraies,
                          Estimée = estimation)
    tab <- knitr::kable(tableau, format = "latex", booktabs = TRUE, digits = 3, row.names = FALSE)
    knitr::asis_output(paste0("\\begin{center}", tab, "\\end{center}"))
  }
}
```

# I Une trajectoire

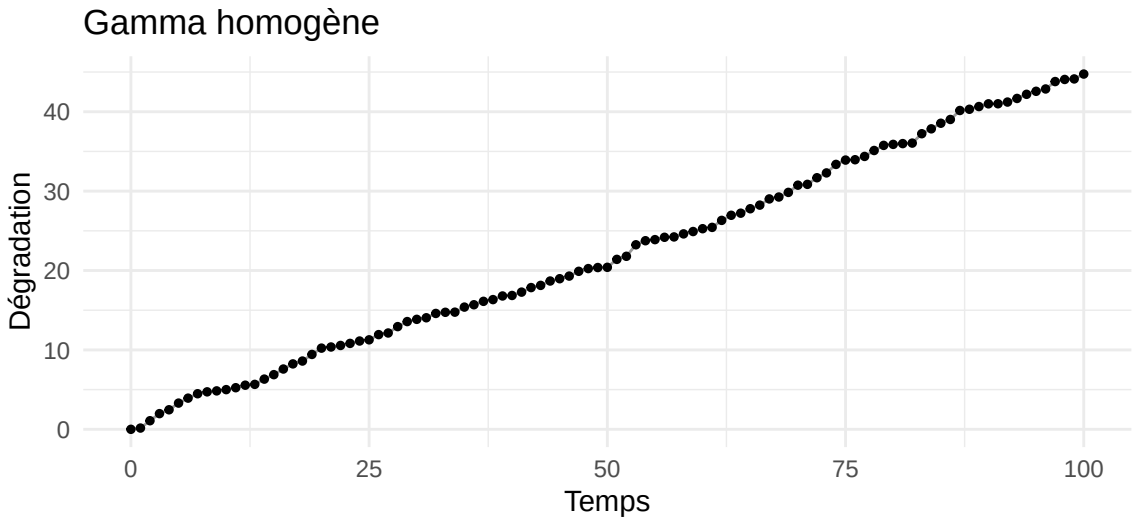
## I.1 Cas 1 : inspections régulières

```
set.seed(1)
t1 <- inspection_times("regular", n = n, tmax = tmax)$time
tracer_plan(t1, "Inspections régulières")
```




### I.1.1 Modèle 1 : gamma homogène

```
x <- sim_gamma(t1)
tracer(t1, x, "Gamma homogène")
```

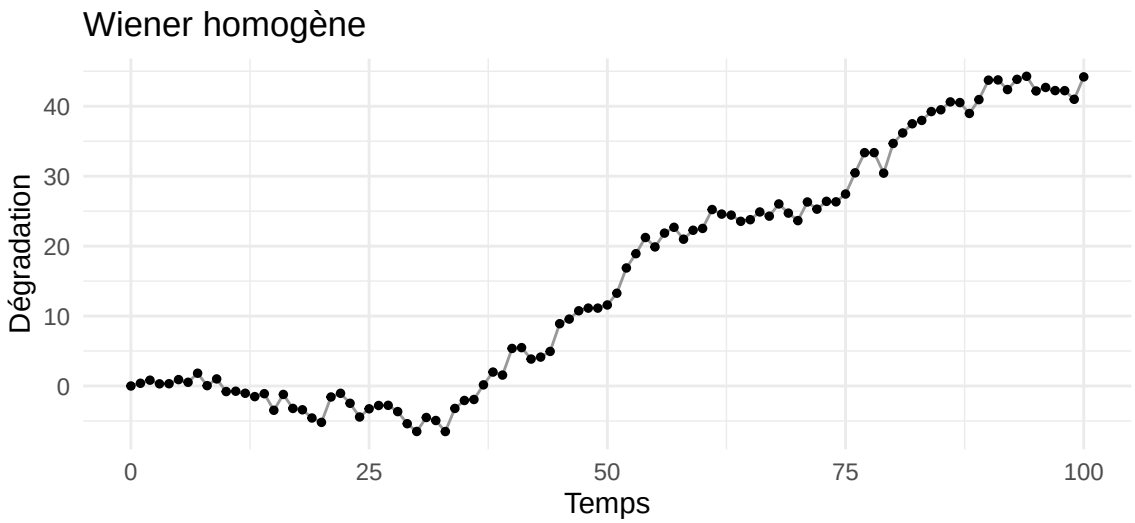



```
resultat(try(gamma_mle(x, t1), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.759 |
| b | 3.0 | 3.932 |

### I.1.2 Modèle 2 : Wiener homogène

```r
x <- sim_wiener(t1)
tracer(t1, x, "Wiener homogène")
```

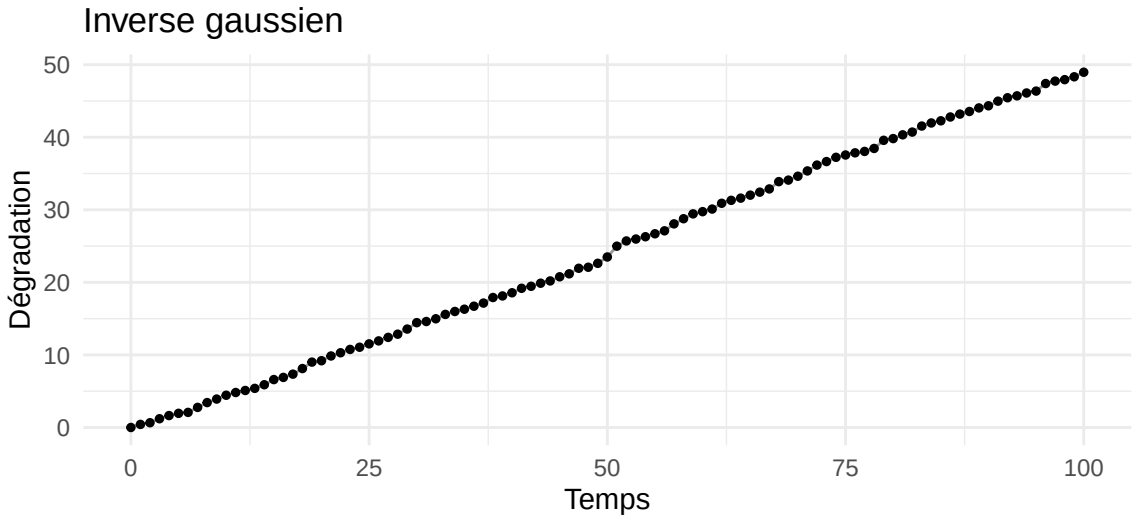



```r
resultat(try(wiener_mle(x, t1), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.442 |
| sigma2 | 2.25 | 2.456 |
| s2 | 2.25 | 2.480 |

### I.1.3 Modèle 3 : inverse gaussien

```r
x <- sim_ig(t1)
tracer(t1, x, "Inverse gaussien")
```

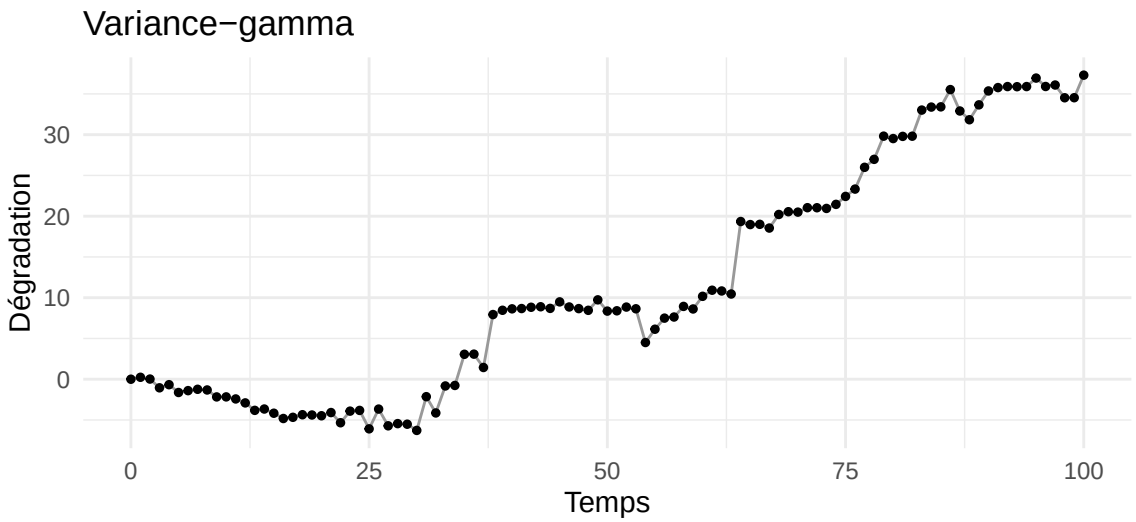



```r
resultat(try(ig_mle(x, t1), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.490 |
| eta | 2.0 | 2.040 |
| eta_unbiased | 2.0 | 1.979 |

### I.1.4 Modèle 4 : variance-gamma

```
x <- sim_vg(t1)
tracer(t1, x, "Variance-gamma")
```

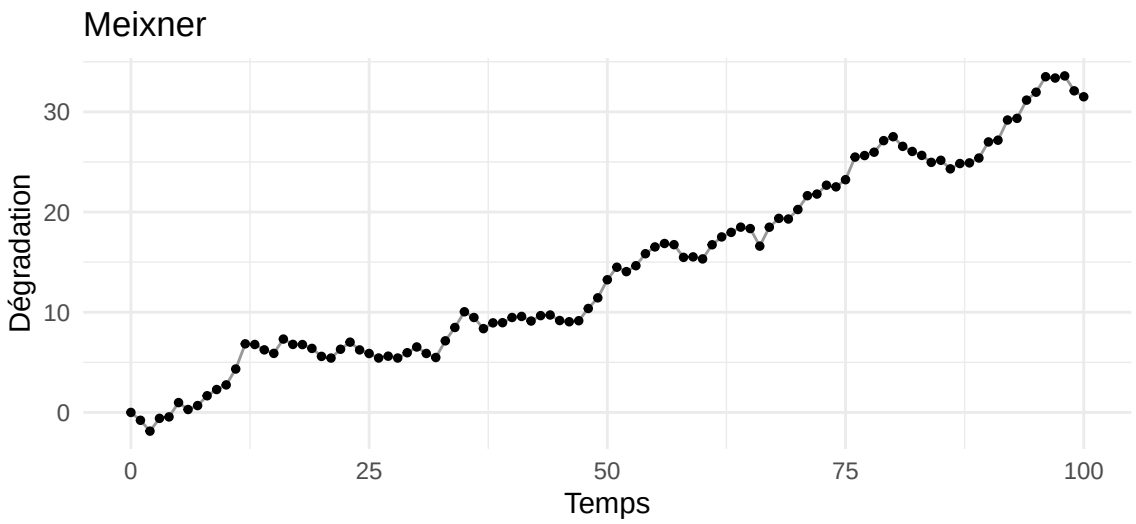



```
resultat(try(vg_mle(x, t1), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.373 |
| sigma | 1.5 | 1.527 |
| nu | 2.0 | 2.351 |

### I.1.5 Modèle 5 : Meixner

```
x <- sim_meixner(t1)
tracer(t1, x, "Meixner")
```

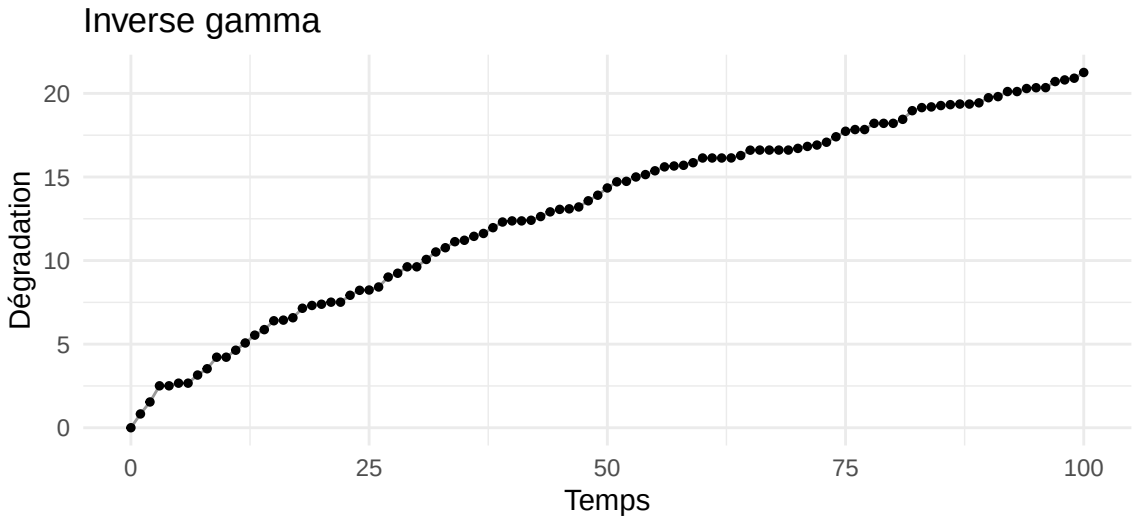



```
resultat(try(meixner_mle(x, t1), silent = TRUE), vrai_meixner)
```

**Estimation impossible** : no interior maximum: all solutions are at the boundary (shifted gamma or Gaussian limit)

### I.1.6 Modèle 6 : inverse gamma

```
x <- sim_igam(t1)
tracer(t1, x, "Inverse gamma")
```

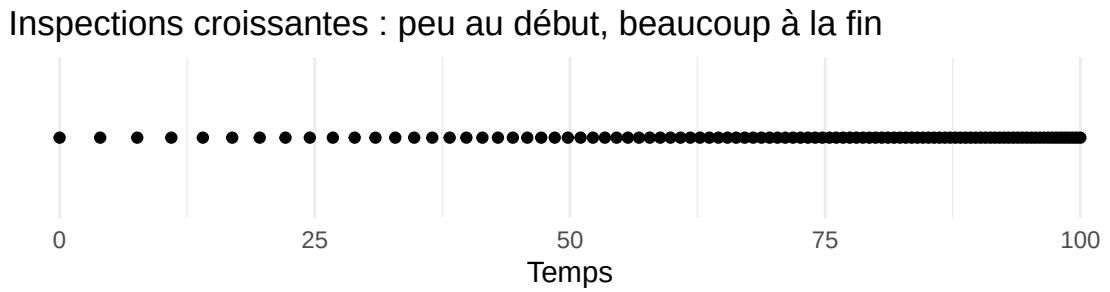



```
resultat(try(igam_mle(x, t1), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 0.958 |
| alpha | 1.0 | 0.946 |
| beta | 1.5 | 1.495 |

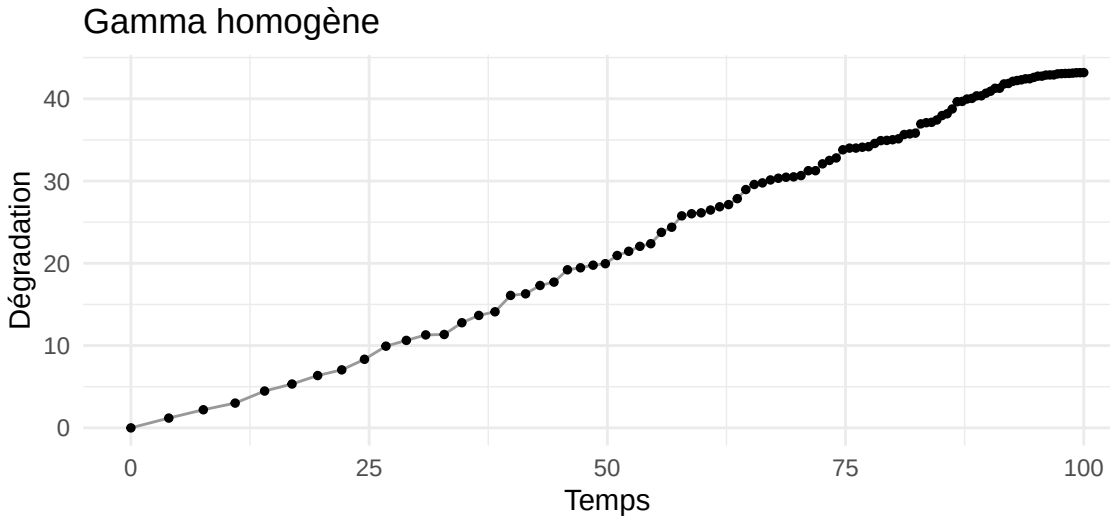
## I.2 Cas 2 : inspections croissantes

```
set.seed(2)
t2 <- inspection_times("increasing", n = n, tmax = tmax)$time
tracer_plan(t2, "Inspections croissantes : peu au début, beaucoup à la fin")
```




### I.2.1 Modèle 1 : gamma homogène

```
x <- sim_gamma(t2)
tracer(t2, x, "Gamma homogène")
```

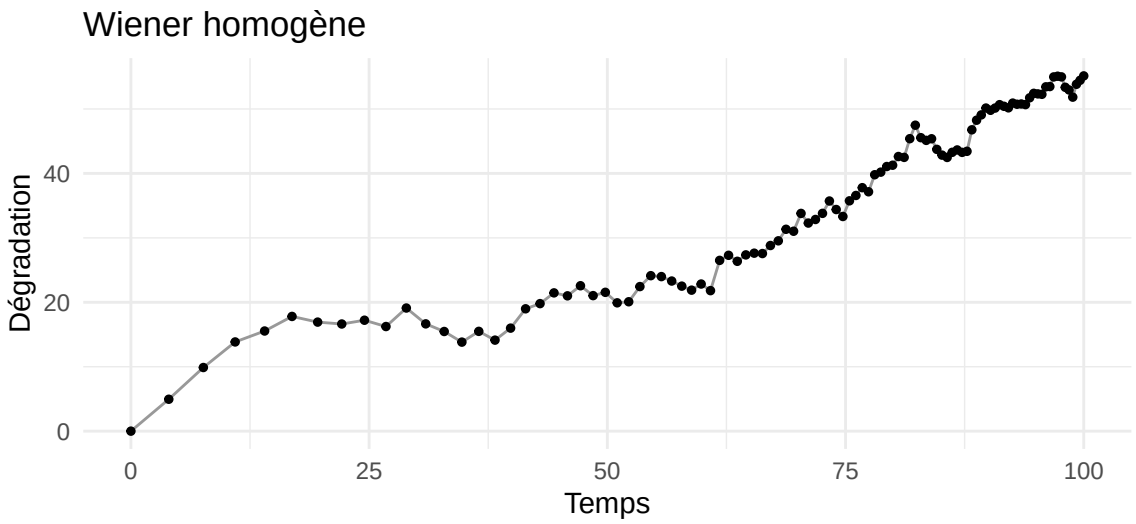



```
resultat(try(gamma_mle(x, t2), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.384 |
| b | 3.0 | 3.204 |

### I.2.2 Modèle 2 : Wiener homogène

```
x <- sim_wiener(t2)
tracer(t2, x, "Wiener homogène")
```

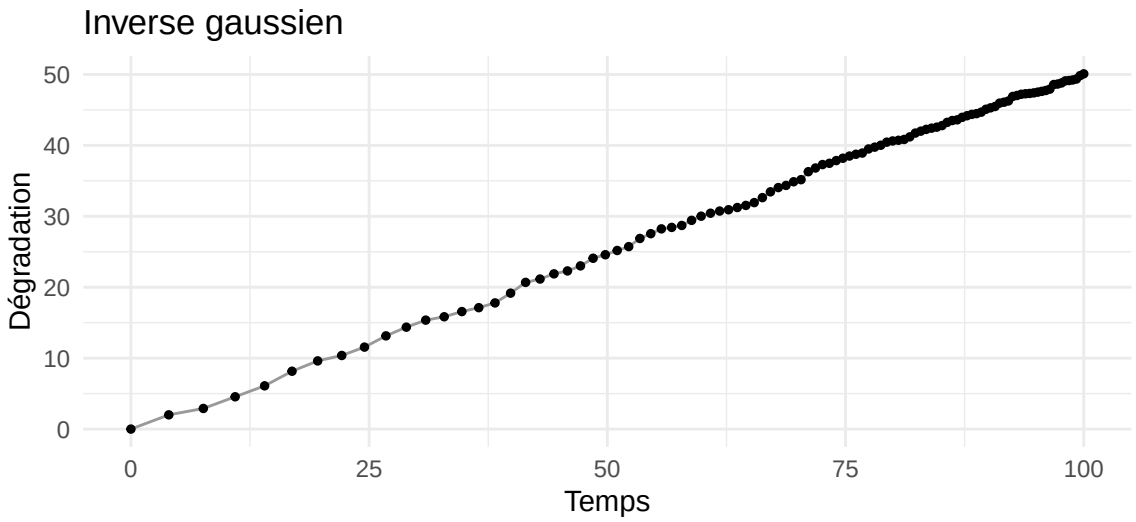



```
resultat(try(wiener_mle(x, t2), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.551 |
| sigma2 | 2.25 | 2.179 |
| s2 | 2.25 | 2.201 |

### I.2.3 Modèle 3 : inverse gaussien

```
x <- sim_ig(t2)
tracer(t2, x, "Inverse gaussien")
```

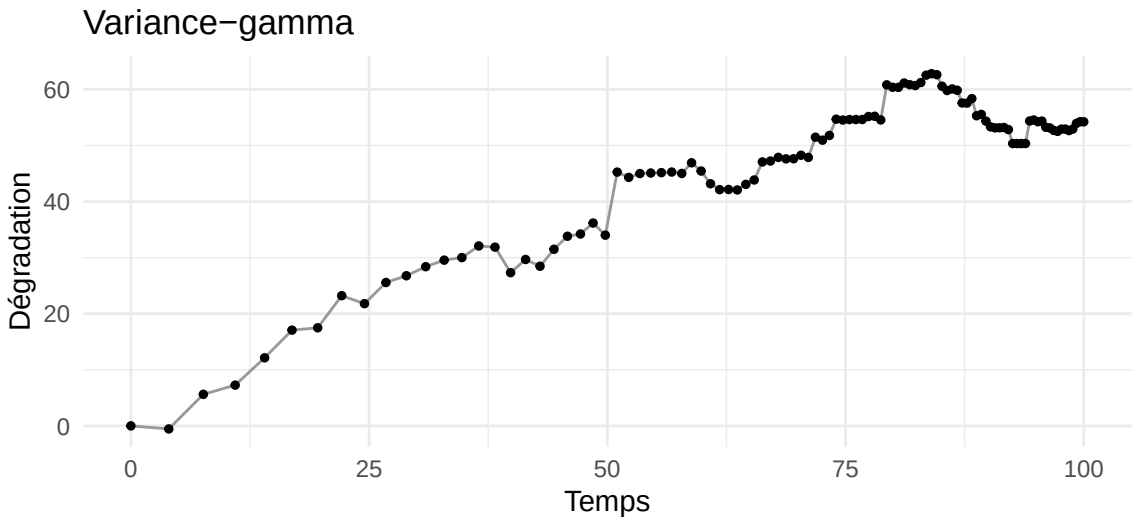


```
resultat(try(ig_mle(x, t2), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.501 |
| eta | 2.0 | 2.019 |
| eta_unbiased | 2.0 | 1.958 |

### I.2.4 Modèle 4 : variance-gamma

```
x <- sim_vg(t2)
tracer(t2, x, "Variance-gamma")
```

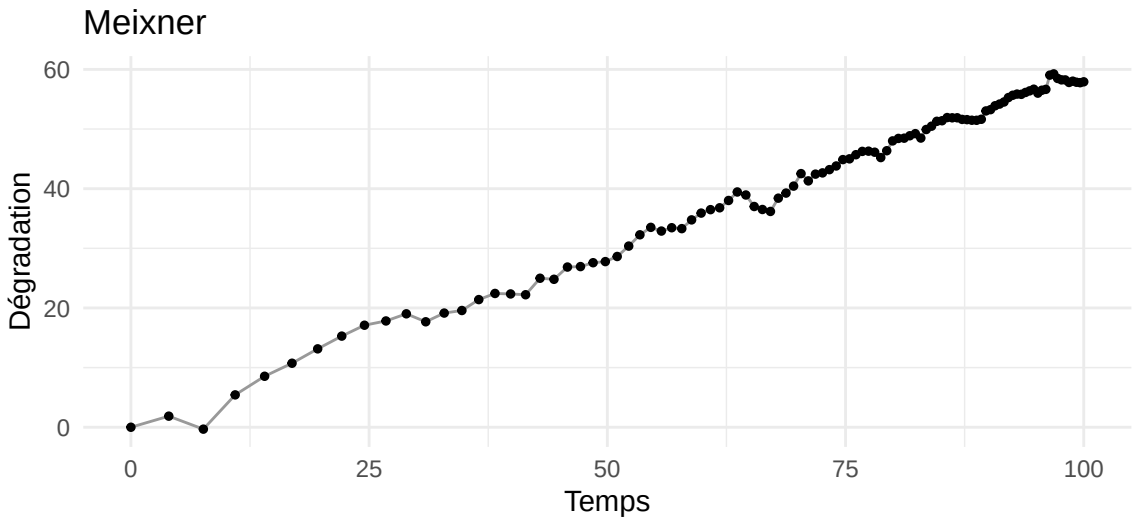



```
resultat(try(vg_mle(x, t2), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.542 |
| sigma | 1.5 | 1.906 |
| nu | 2.0 | 1.855 |

### I.2.5 Modèle 5 : Meixner

```
x <- sim_meixner(t2)
tracer(t2, x, "Meixner")
```

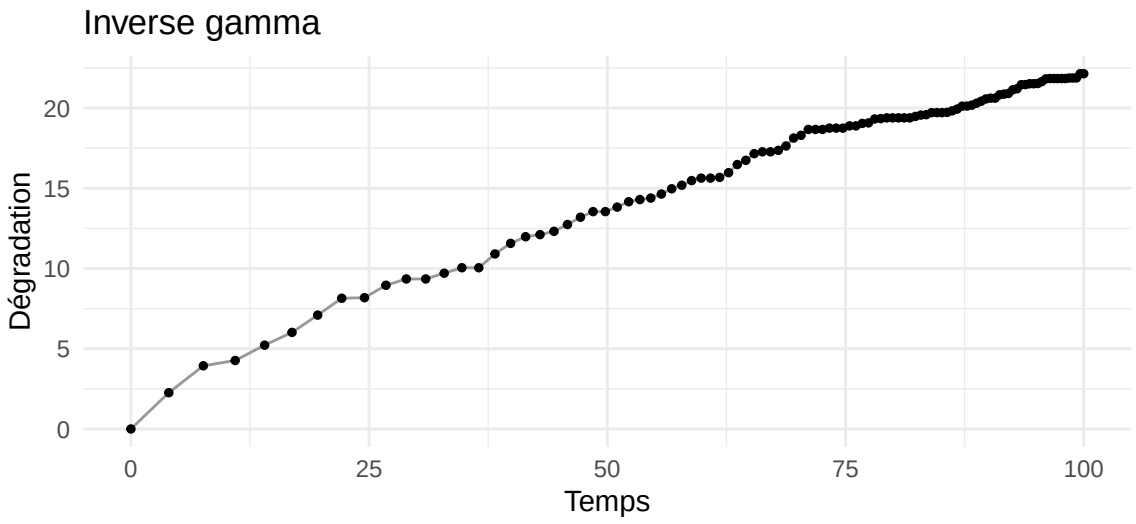



```
resultat(try(meixner_mle(x, t2), silent = TRUE), vrai_meixner)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.0 | 1.936 |
| b | 0.3 | 0.276 |
| d | 1.5 | 0.480 |
| m | 0.2 | 0.450 |

#### I.2.6 Modèle 6 : inverse gamma

```
x <- sim_igam(t2)
tracer(t2, x, "Inverse gamma")
```

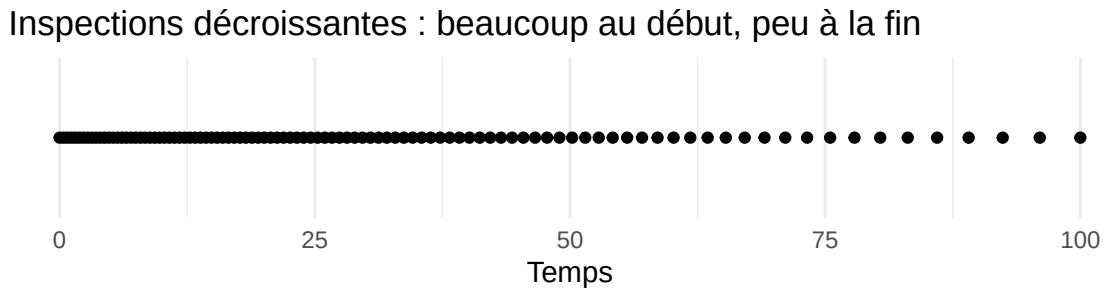


```
resultat(try(igam_mle(x, t2), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 1.001 |
| alpha | 1.0 | 0.913 |
| beta | 1.5 | 1.446 |

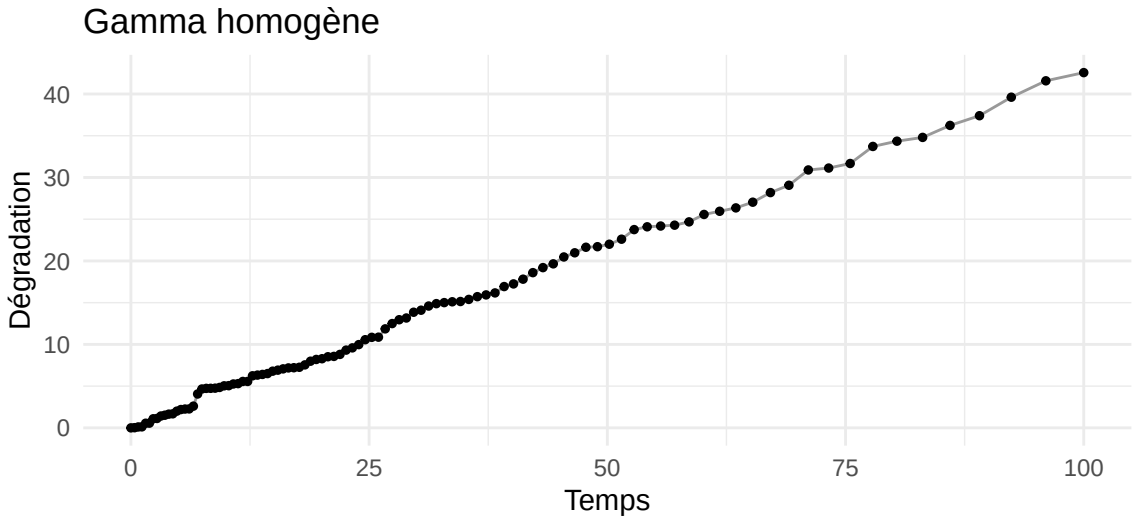
## I.3 Cas 3 : inspections décroissantes

```
set.seed(3)
t3 <- inspection_times("decreasing", n = n, tmax = tmax)$time
tracer_plan(t3, "Inspections décroissantes : beaucoup au début, peu à la fin")
```




### I.3.1 Modèle 1 : gamma homogène

```
x <- sim_gamma(t3)
tracer(t3, x, "Gamma homogène")
```

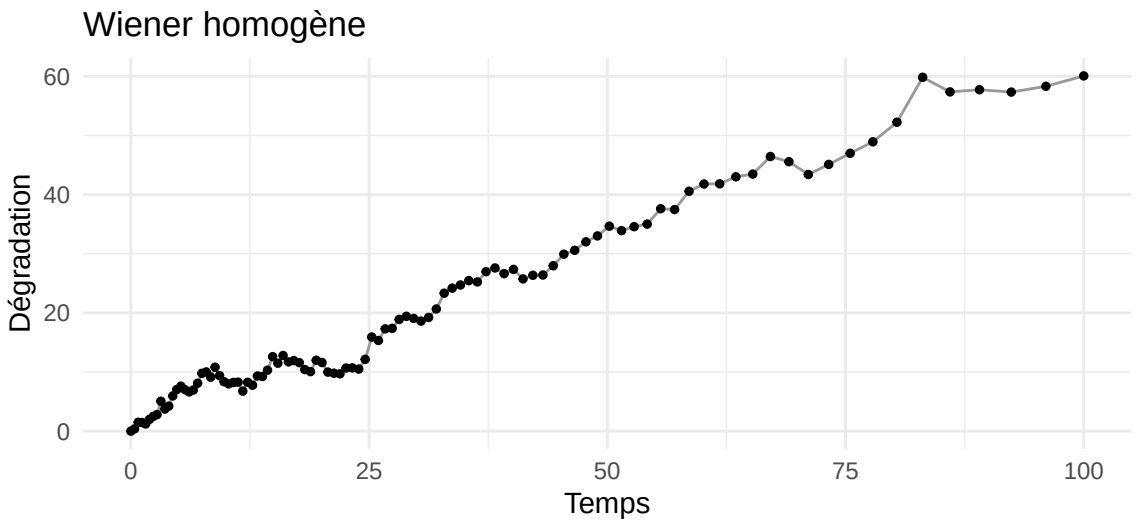



```
resultat(try(gamma_mle(x, t3), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.683 |
| b | 3.0 | 3.955 |

### I.3.2 Modèle 2 : Wiener homogène

```
x <- sim_wiener(t3)
tracer(t3, x, "Wiener homogène")
```

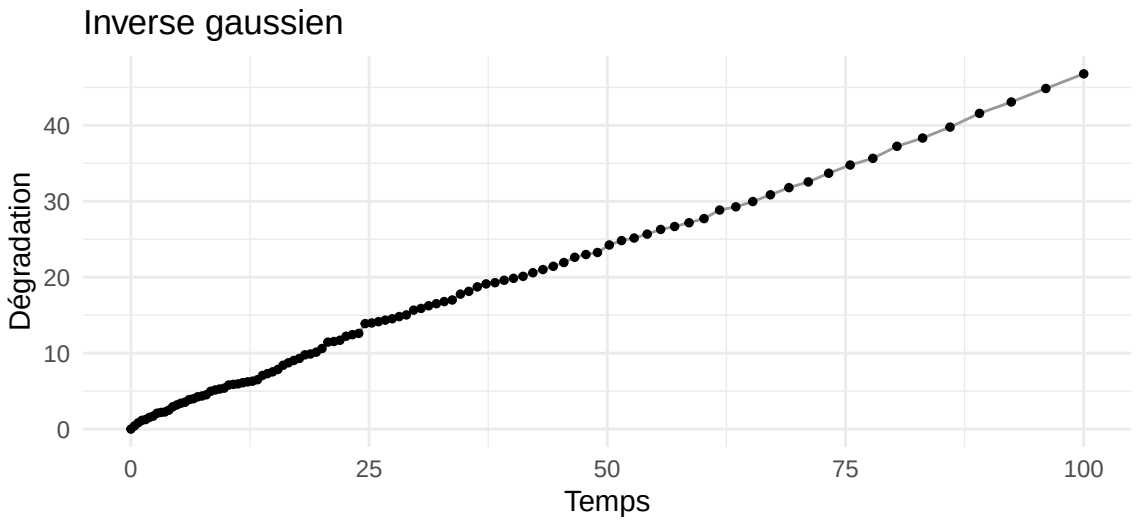



```
resultat(try(wiener_mle(x, t3), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.601 |
| sigma2 | 2.25 | 1.939 |
| s2 | 2.25 | 1.958 |

### I.3.3 Modèle 3 : inverse gaussien

```
x <- sim_ig(t3)
tracer(t3, x, "Inverse gaussien")
```

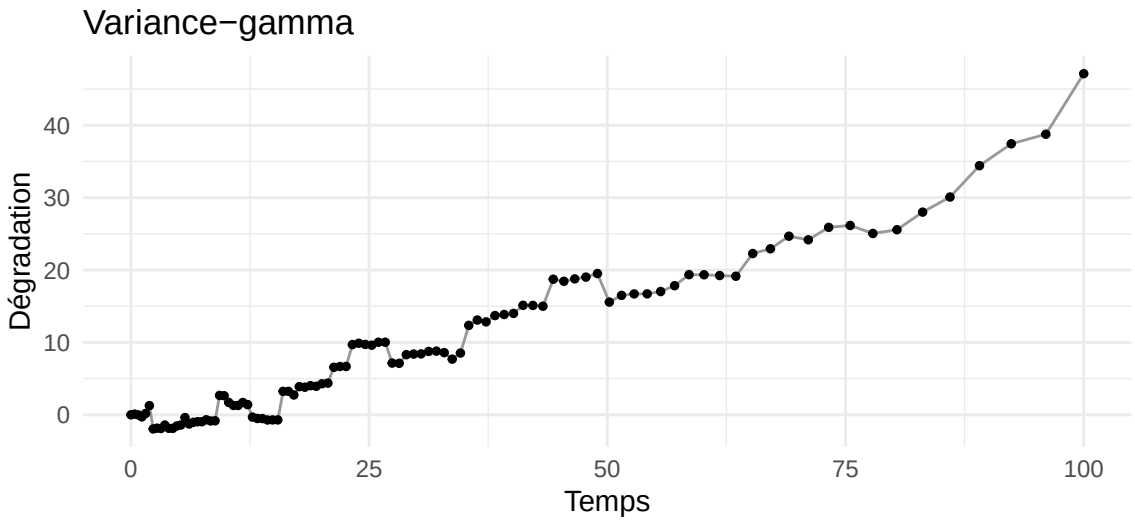


```r
resultat(try(ig_mle(x, t3), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.468 |
| eta | 2.0 | 2.091 |
| eta_unbiased | 2.0 | 2.029 |

### I.3.4 Modèle 4 : variance-gamma

```r
x <- sim_vg(t3)
tracer(t3, x, "Variance-gamma")
```

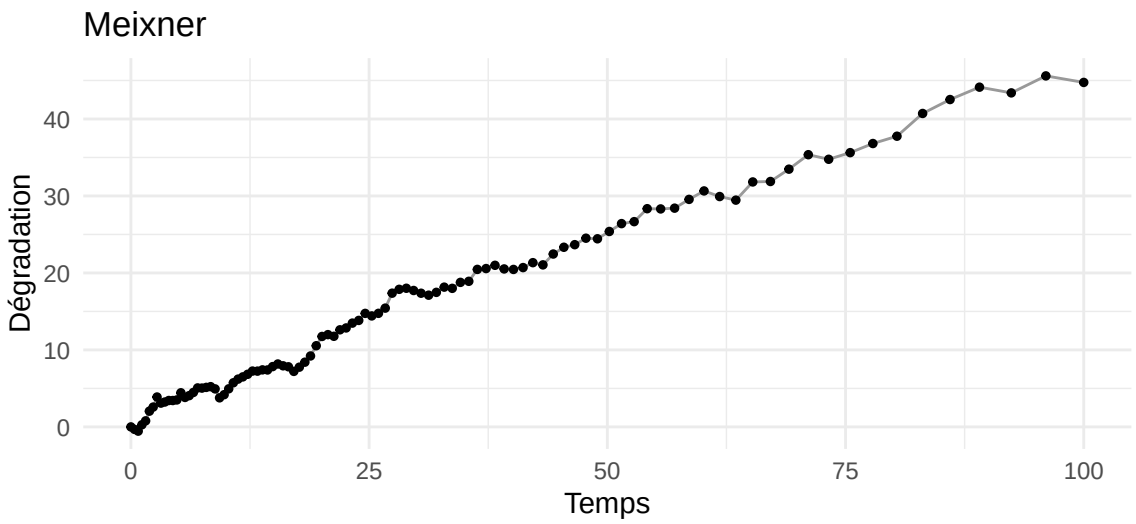



```r
resultat(try(vg_mle(x, t3), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.471 |
| sigma | 1.5 | 1.290 |
| nu | 2.0 | 2.134 |

### I.3.5 Modèle 5 : Meixner

```r
x <- sim_meixner(t3)
tracer(t3, x, "Meixner")
```

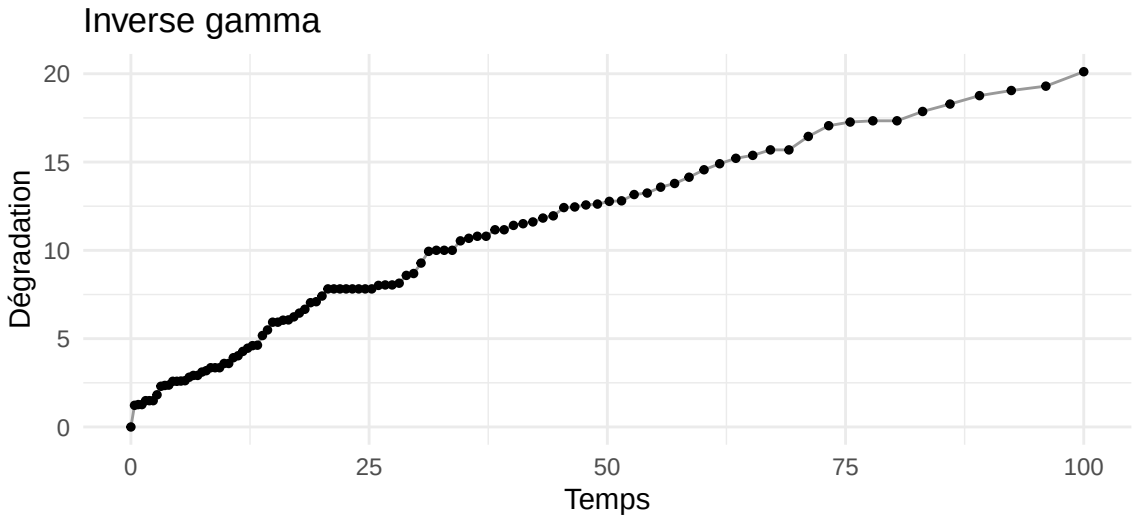



```
resultat(try(meixner_mle(x, t3), silent = TRUE), vrai_meixner)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.0 | 0.603 |
| b | 0.3 | 0.121 |
| d | 1.5 | 2.844 |
| m | 0.2 | 0.343 |

#### I.3.6 Modèle 6 : inverse gamma

```
x <- sim_igam(t3)
tracer(t3, x, "Inverse gamma")
```

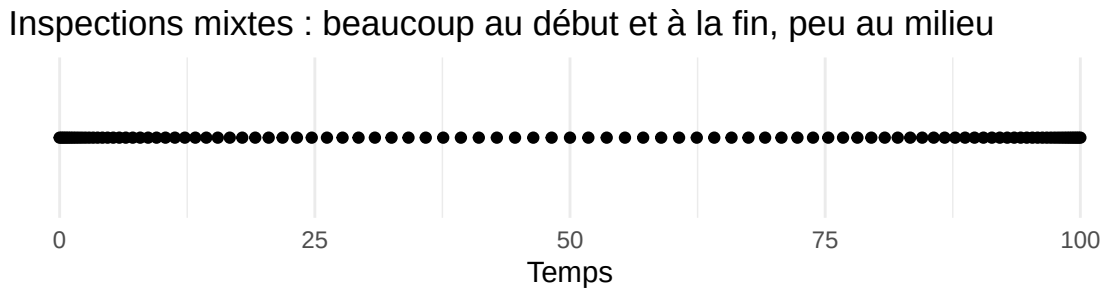


```
resultat(try(igam_mle(x, t3), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 1.075 |
| alpha | 1.0 | 0.867 |
| beta | 1.5 | 1.443 |

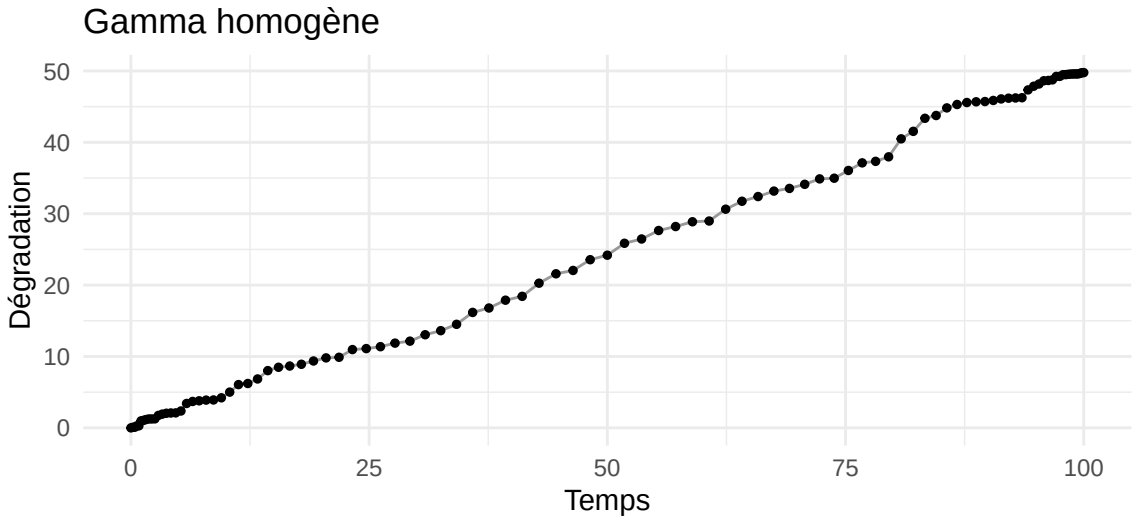
## I.4 Cas 4 : inspections mixtes

```
set.seed(4)
t4 <- inspection_times("dec-inc", n = n, tmax = tmax)$time
tracer_plan(t4, "Inspections mixtes : beaucoup au début et à la fin, peu au milieu")
```




### I.4.1 Modèle 1 : gamma homogène

```
x <- sim_gamma(t4)
tracer(t4, x, "Gamma homogène")
```

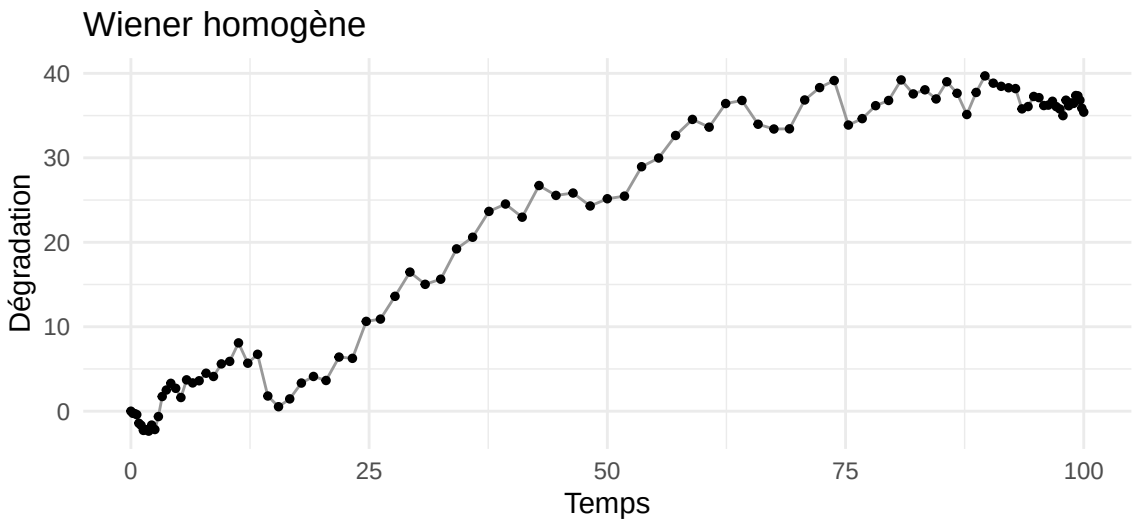



```
resultat(try(gamma_mle(x, t4), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.357 |
| b | 3.0 | 2.727 |

### I.4.2 Modèle 2 : Wiener homogène

```
x <- sim_wiener(t4)
tracer(t4, x, "Wiener homogène")
```

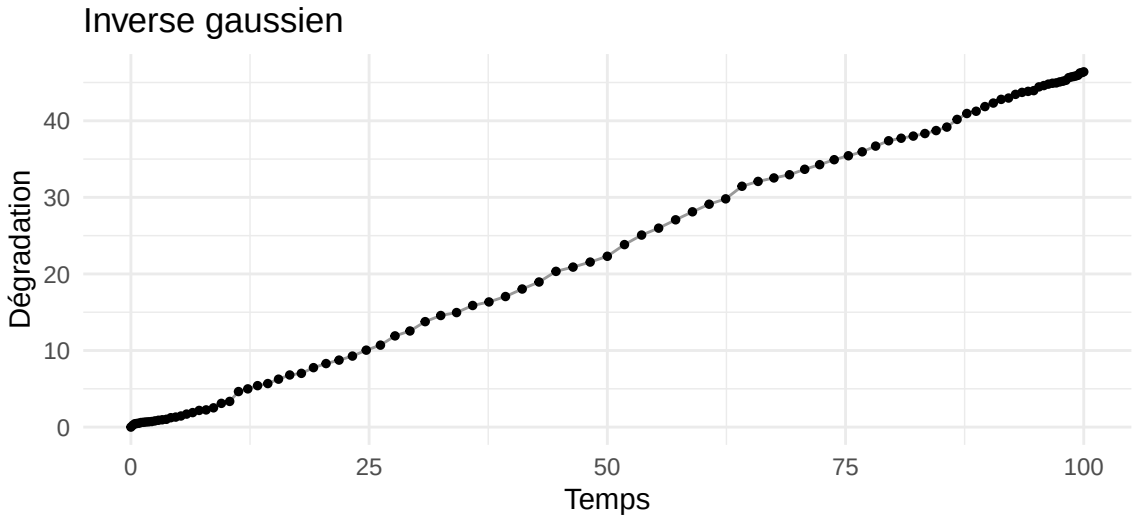



```
resultat(try(wiener_mle(x, t4), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.354 |
| sigma2 | 2.25 | 2.571 |
| s2 | 2.25 | 2.597 |

### I.4.3 Modèle 3 : inverse gaussien

```
x <- sim_ig(t4)
tracer(t4, x, "Inverse gaussien")
```

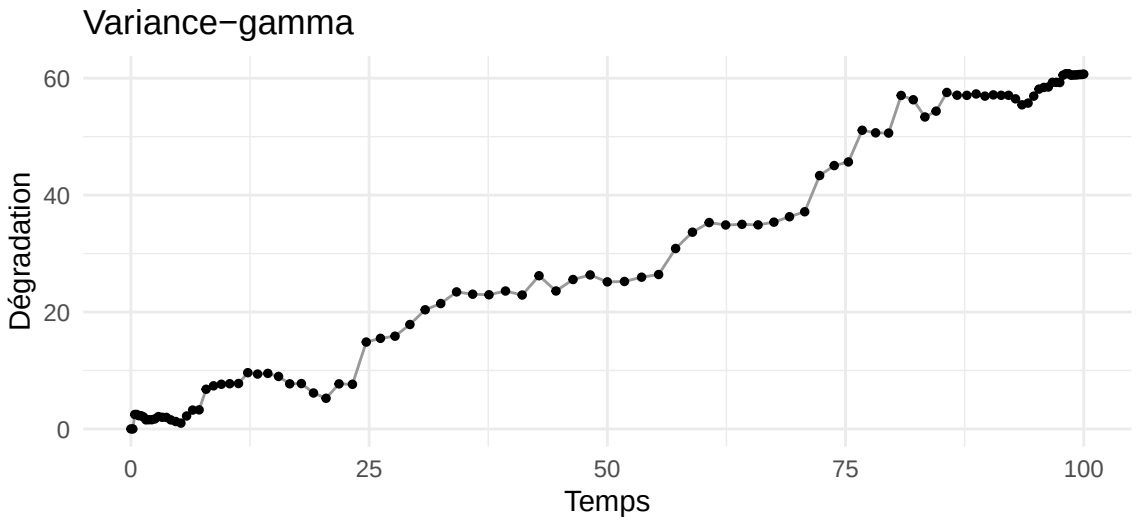


```
resultat(try(ig_mle(x, t4), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.464 |
| eta | 2.0 | 1.781 |
| eta_unbiased | 2.0 | 1.728 |

### I.4.4 Modèle 4 : variance-gamma

```
x <- sim_vg(t4)
tracer(t4, x, "Variance-gamma")
```

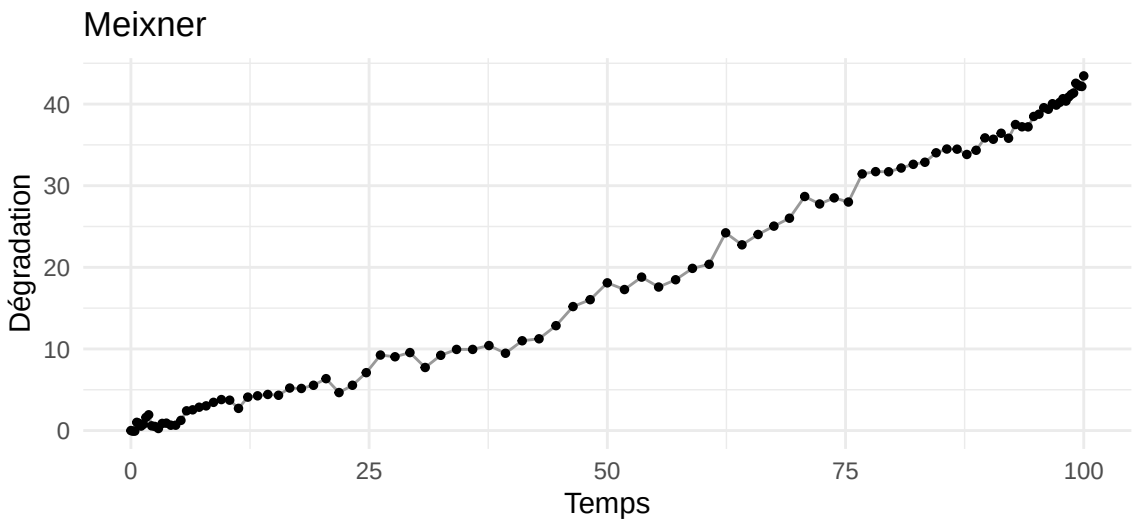



```
resultat(try(vg_mle(x, t4), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.607 |
| sigma | 1.5 | 1.375 |
| nu | 2.0 | 2.012 |

### I.4.5 Modèle 5 : Meixner

```
x <- sim_meixner(t4)
tracer(t4, x, "Meixner")
```

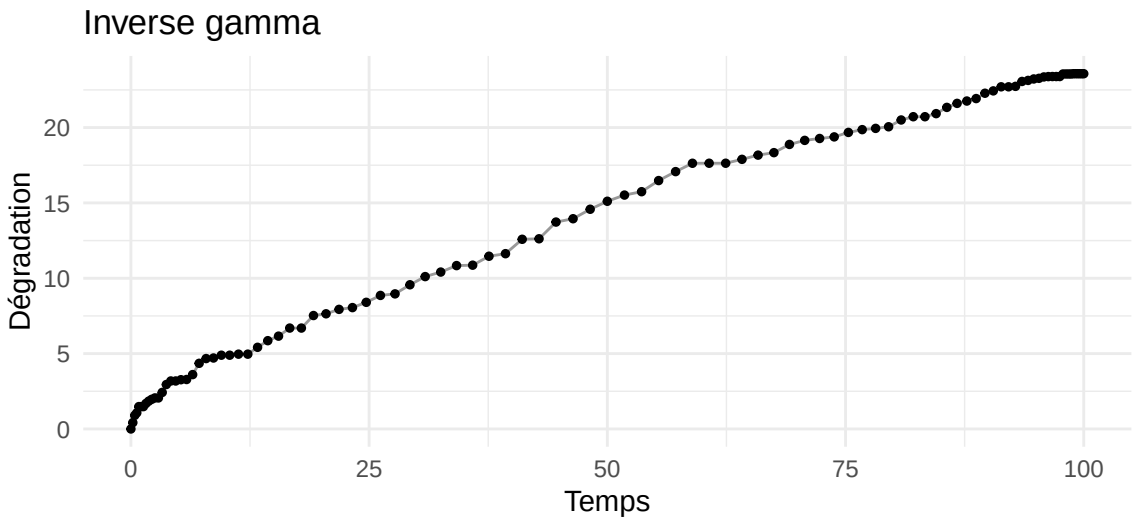



```
resultat(try(meixner_mle(x, t4), silent = TRUE), vrai_meixner)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.0 | 1.165 |
| b | 0.3 | 0.732 |
| d | 1.5 | 1.105 |
| m | 0.2 | -0.059 |

#### I.4.6 Modèle 6 : inverse gamma

```
x <- sim_igam(t4)
tracer(t4, x, "Inverse gamma")
```



```
resultat(try(igam_mle(x, t4), silent = TRUE), vrai_igam)
```

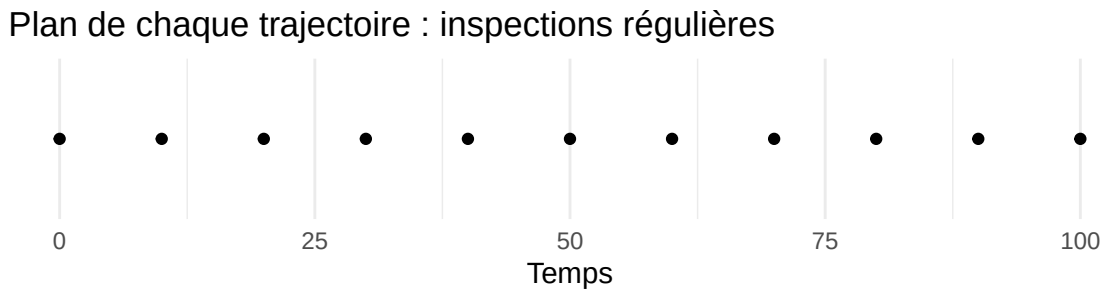
| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 0.863 |
| alpha | 1.0 | 0.774 |
| beta | 1.5 | 1.393 |

# II Plusieurs trajectoires

## II.1 Cas 5 : inspections régulières

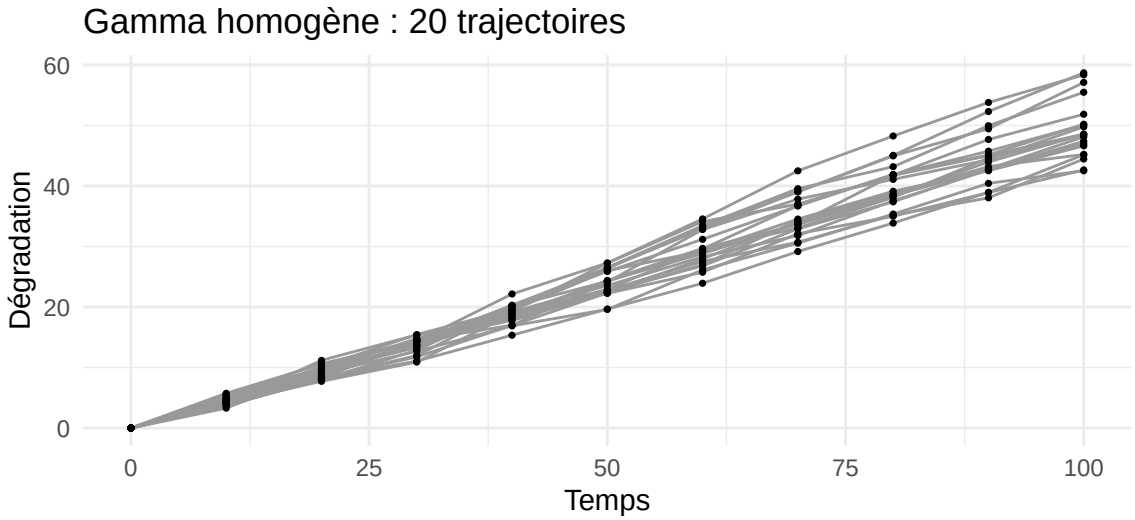
20 trajectoires indépendantes, chacune inspectée 10 fois jusqu'à 100. Chaque trajectoire est un `data.frame(t, X)`, et les fonctions du package reçoivent la liste de ces tableaux.

```
set.seed(5)
t5 <- inspection_times("regular", n = 10, tmax = tmax)$time
tracer_plan(t5, "Plan de chaque trajectoire : inspections régulières")
```




### II.1.1 Modèle 1 : gamma homogène

```
liste <- simuler(sim_gamma, t5)
tracer_plusieurs(liste, paste("Gamma homogène :", n_traj, "trajectoires"))
```

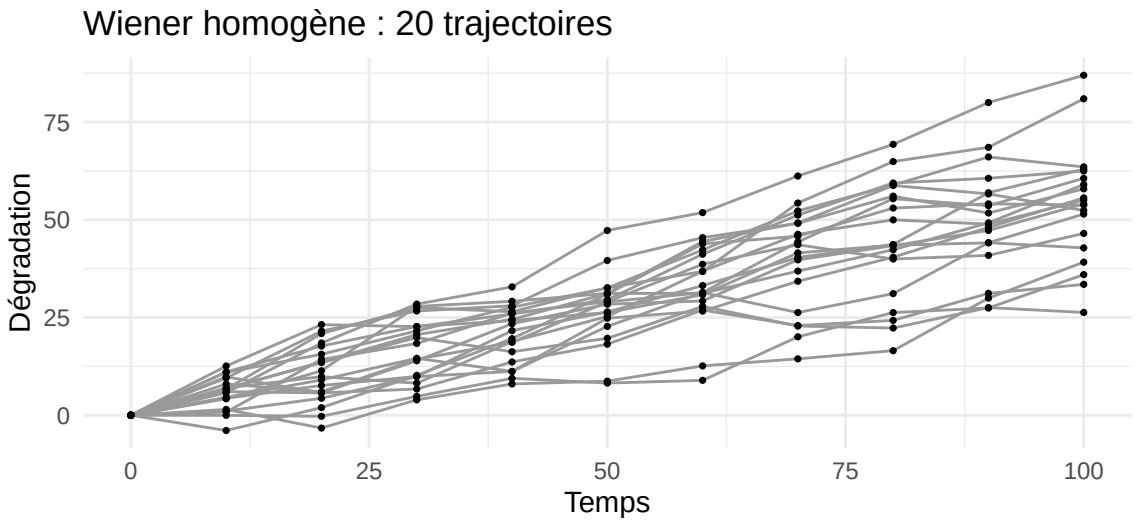



```
resultat(try(gamma_mle(liste), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.522 |
| b | 3.0 | 3.086 |

### II.1.2 Modèle 2 : Wiener homogène

```
liste <- simuler(sim_wiener, t5)
tracer_plusieurs(liste, paste("Wiener homogène :", n_traj, "trajectoires"))
```

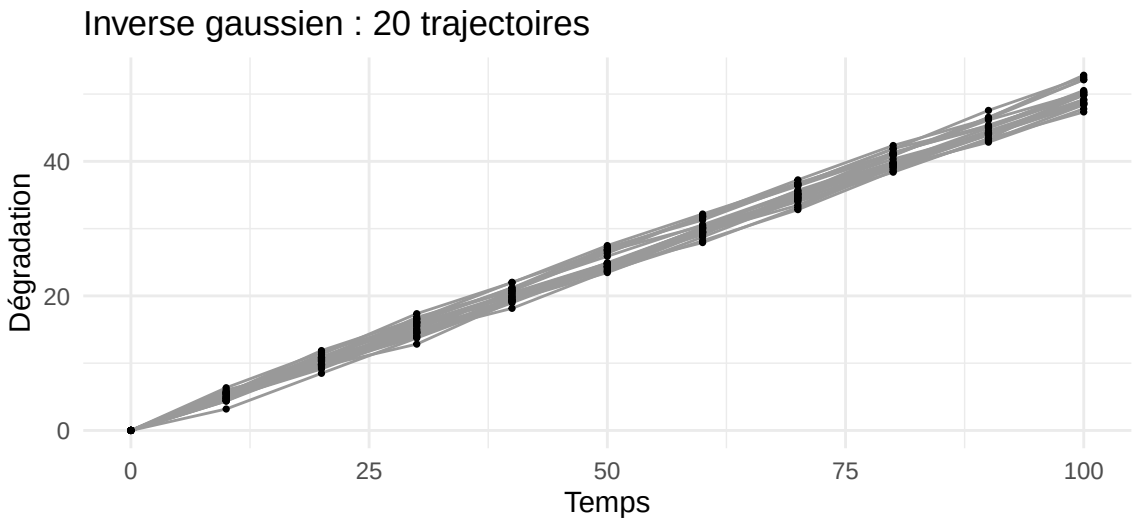



```
resultat(try(wiener_mle(liste), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.541 |
| sigma2 | 2.25 | 2.167 |
| s2 | 2.25 | 2.178 |

### II.1.3 Modèle 3 : inverse gaussien

```
liste <- simuler(sim_ig, t5)
tracer_plusieurs(liste, paste("Inverse gaussien :", n_traj, "trajectoires"))
```

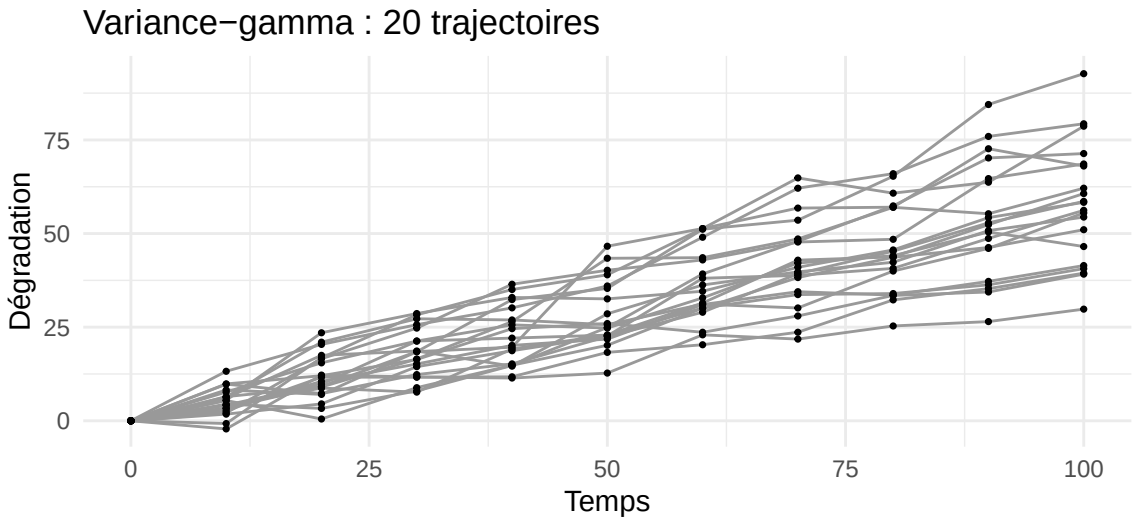


```
resultat(try(ig_mle(liste), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.498 |
| eta | 2.0 | 2.327 |
| eta_unbiased | 2.0 | 2.292 |

### II.1.4 Modèle 4 : variance-gamma

```
liste <- simuler(sim_vg, t5)
tracer_plusieurs(liste, paste("Variance-gamma :", n_traj, "trajectoires"))
```

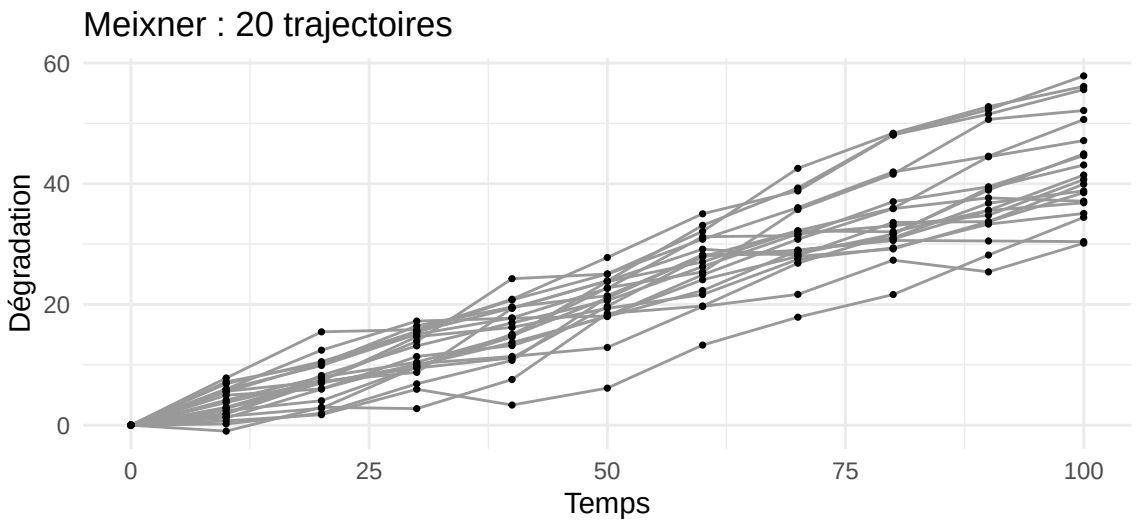



```
resultat(try(vg_mle(liste), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.576 |
| sigma | 1.5 | 1.377 |
| nu | 2.0 | 1.844 |

### II.1.5 Modèle 5 : Meixner

```
liste <- simuler(sim_meixner, t5)
tracer_plusieurs(liste, paste("Meixner :", n_traj, "trajectoires"))
```

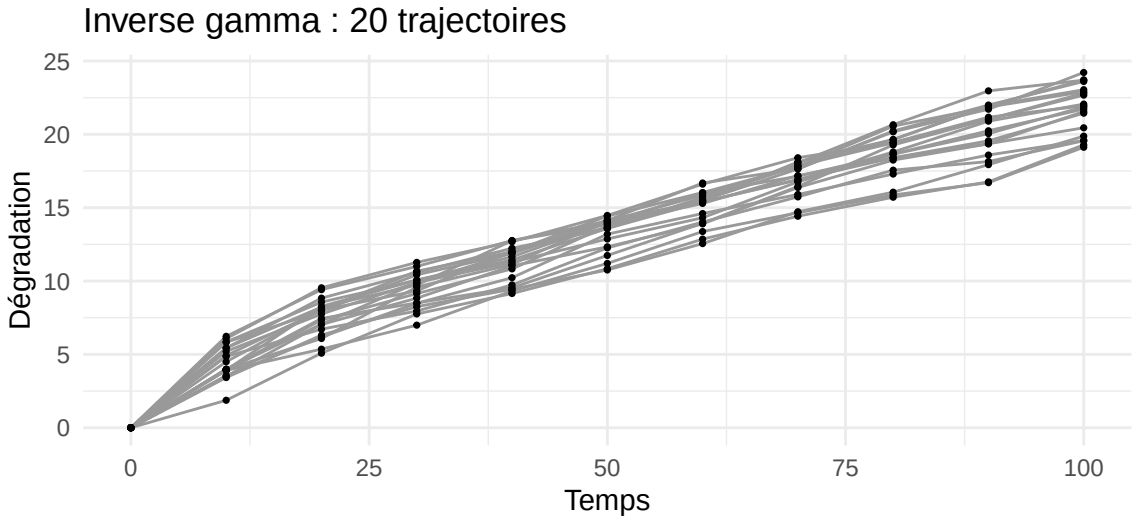



```
resultat(try(meixner_mle(liste), silent = TRUE), vrai_meixner)
```

**Estimation impossible** : no interior maximum: all solutions are at the boundary (shifted gamma or Gaussian limit)

#### II.1.6 Modèle 6 : inverse gamma

```
liste <- simuler(sim_igam, t5)
tracer_plusieurs(liste, paste("Inverse gamma :", n_traj, "trajectoires"))
```



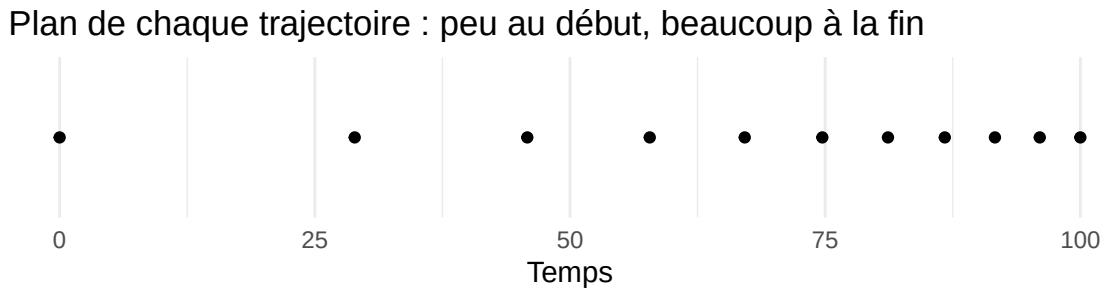

```
resultat(try(igam_mle(liste), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 0.967 |
| alpha | 1.0 | 0.843 |
| beta | 1.5 | 1.428 |

## II.2 Cas 6 : inspections croissantes

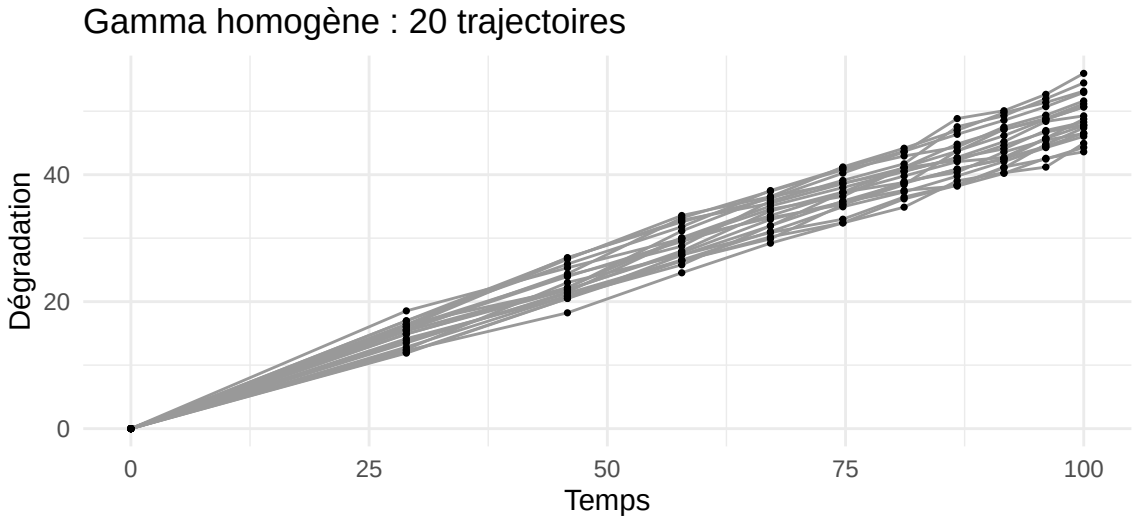
20 trajectoires indépendantes, chacune inspectée 10 fois jusqu'à 100. Chaque trajectoire est un `data.frame(t, X)`, et les fonctions du package reçoivent la liste de ces tableaux.

```
set.seed(6)
t6 <- inspection_times("increasing", n = 10, tmax = tmax)$time
tracer_plan(t6, "Plan de chaque trajectoire : peu au début, beaucoup à la fin")
```




### II.2.1 Modèle 1 : gamma homogène

```
liste <- simuler(sim_gamma, t6)
tracer_plusieurs(liste, paste("Gamma homogène :", n_traj, "trajectoires"))
```

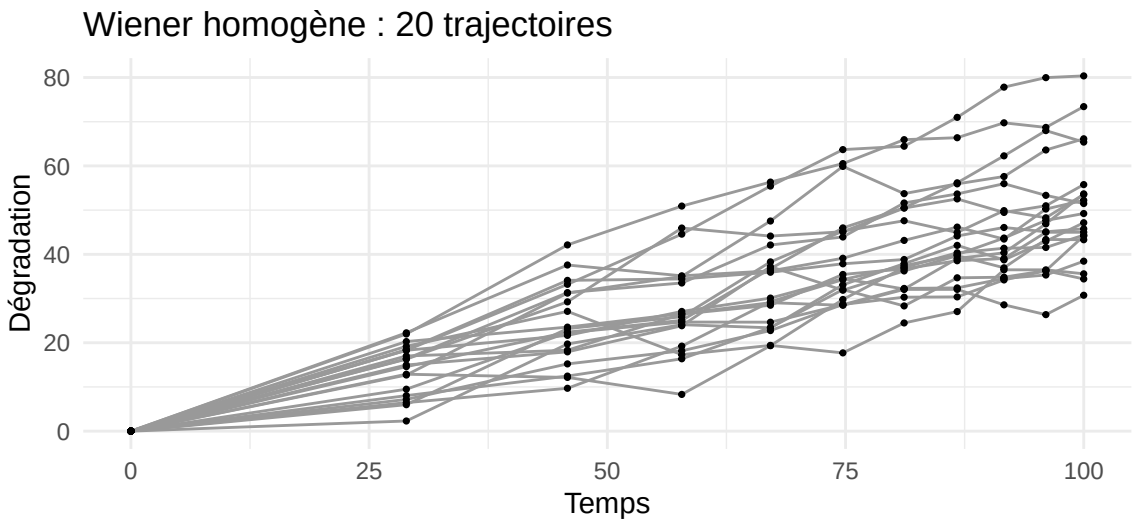



```
resultat(try(gamma_mle(liste), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.522 |
| b | 3.0 | 3.101 |

### II.2.2 Modèle 2 : Wiener homogène

```
liste <- simuler(sim_wiener, t6)
tracer_plusieurs(liste, paste("Wiener homogène :", n_traj, "trajectoires"))
```

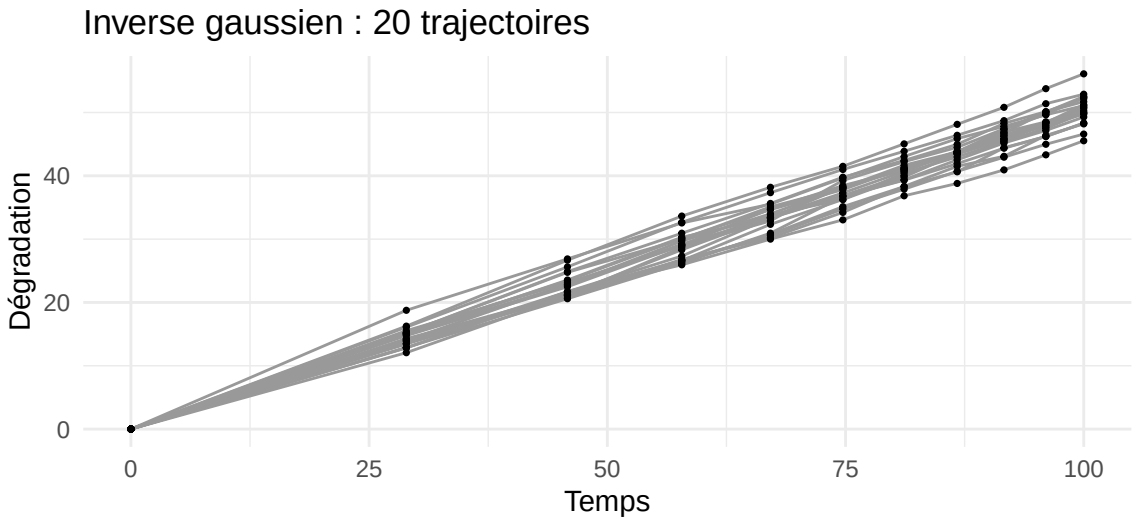



```
resultat(try(wiener_mle(liste), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.505 |
| sigma2 | 2.25 | 2.200 |
| s2 | 2.25 | 2.211 |

### II.2.3 Modèle 3 : inverse gaussien

```
liste <- simuler(sim_ig, t6)
tracer_plusieurs(liste, paste("Inverse gaussien :", n_traj, "trajectoires"))
```

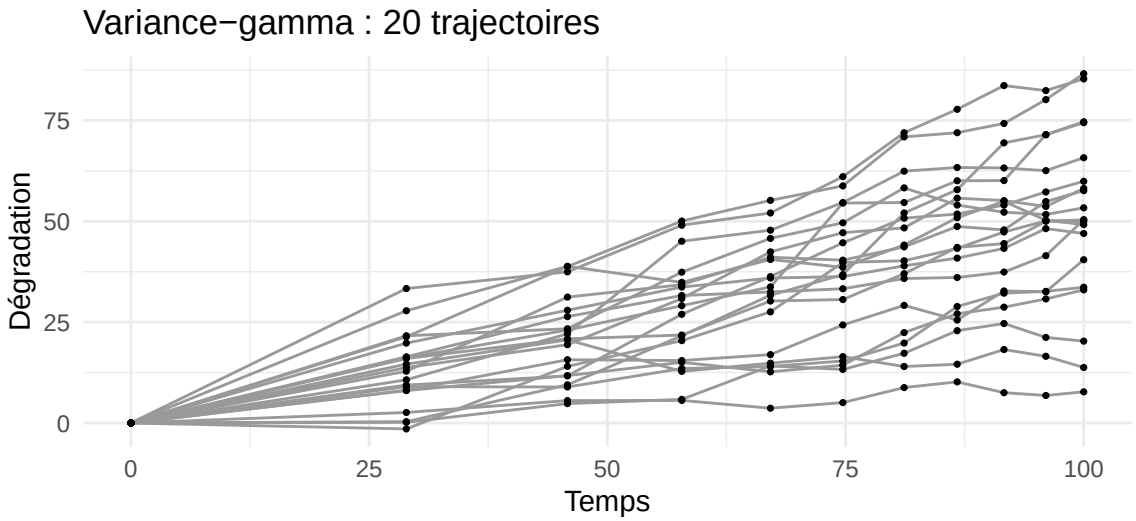


```r
resultat(try(ig_mle(liste), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.504 |
| eta | 2.0 | 2.102 |
| eta_unbiased | 2.0 | 2.070 |

### II.2.4 Modèle 4 : variance-gamma

```r
liste <- simuler(sim_vg, t6)
tracer_plusieurs(liste, paste("Variance-gamma :", n_traj, "trajectoires"))
```

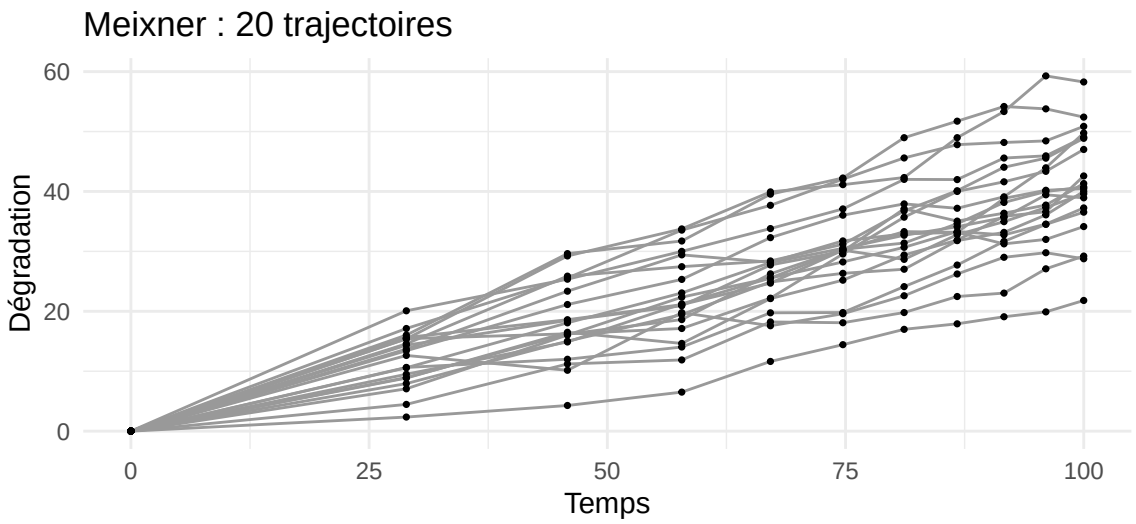



```r
resultat(try(vg_mle(liste), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.506 |
| sigma | 1.5 | 1.494 |
| nu | 2.0 | 1.659 |

### II.2.5 Modèle 5 : Meixner

```r
liste <- simuler(sim_meixner, t6)
tracer_plusieurs(liste, paste("Meixner :", n_traj, "trajectoires"))
```

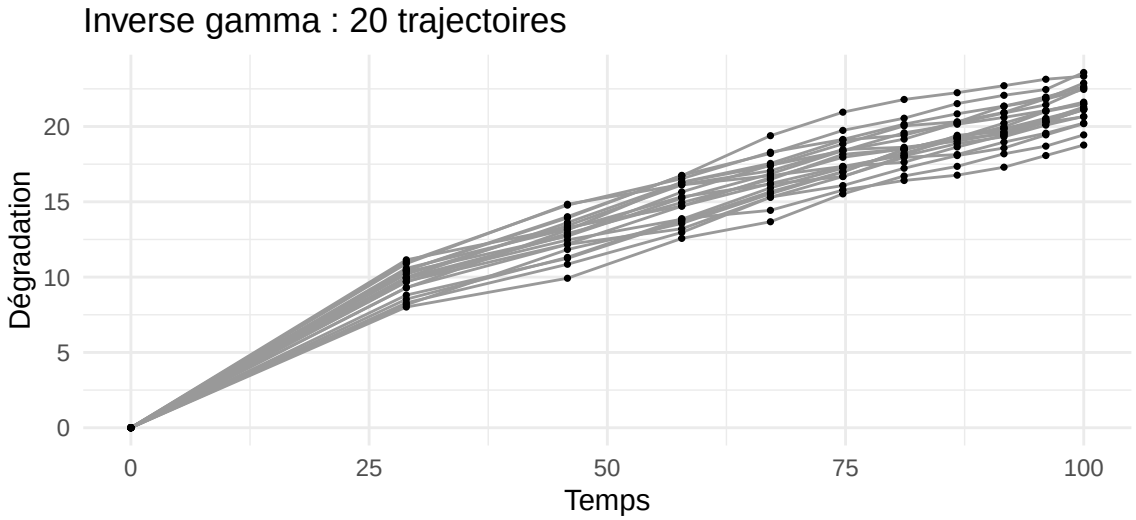



```
resultat(try(meixner_mle(liste), silent = TRUE), vrai_meixner)
```

**Estimation impossible** : no interior maximum: all solutions are at the boundary (shifted gamma or Gaussian limit)

### II.2.6 Modèle 6 : inverse gamma

```
liste <- simuler(sim_igam, t6)
tracer_plusieurs(liste, paste("Inverse gamma :", n_traj, "trajectoires"))
```



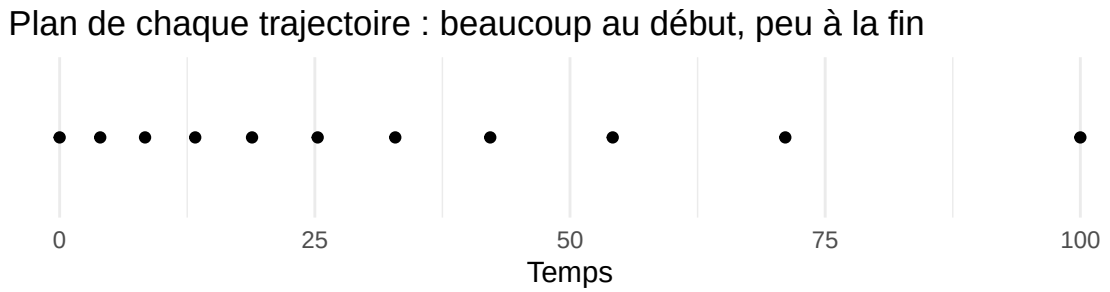

```
resultat(try(igam_mle(liste), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 0.888 |
| alpha | 1.0 | 1.009 |
| beta | 1.5 | 1.548 |

## II.3 Cas 7 : inspections décroissantes

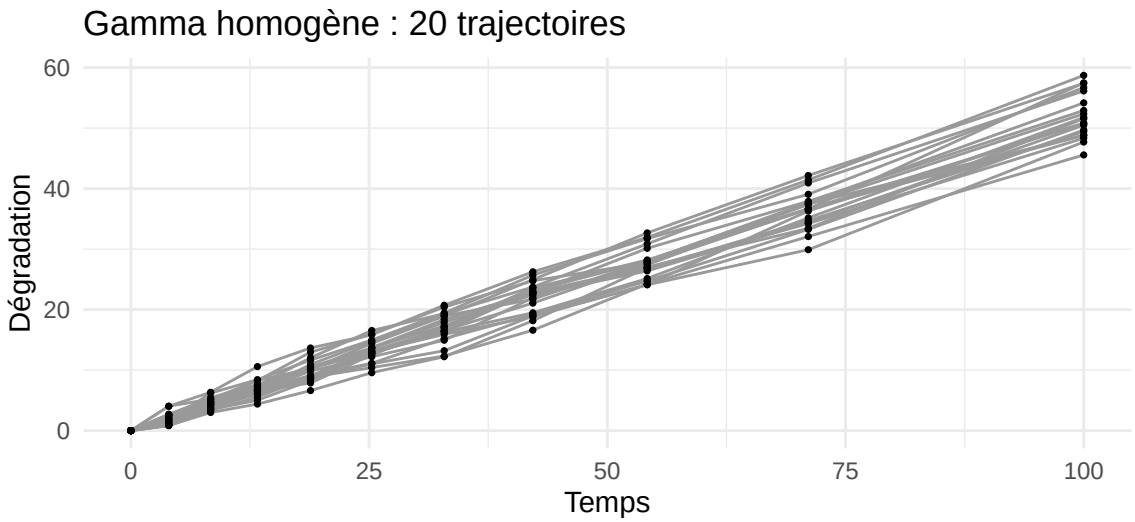
20 trajectoires indépendantes, chacune inspectée 10 fois jusqu'à 100. Chaque trajectoire est un `data.frame(t, X)`, et les fonctions du package reçoivent la liste de ces tableaux.

```
set.seed(7)
t7 <- inspection_times("decreasing", n = 10, tmax = tmax)$time
tracer_plan(t7, "Plan de chaque trajectoire : beaucoup au début, peu à la fin")
```




### II.3.1 Modèle 1 : gamma homogène

```
liste <- simuler(sim_gamma, t7)
tracer_plusieurs(liste, paste("Gamma homogène :", n_traj, "trajectoires"))
```

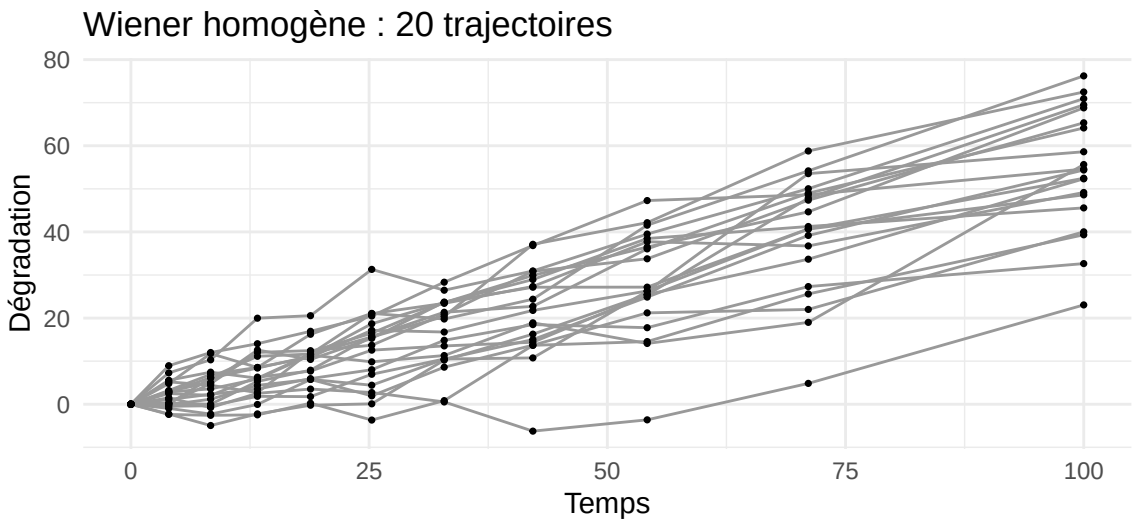



```
resultat(try(gamma_mle(liste), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.725 |
| b | 3.0 | 3.319 |

### II.3.2 Modèle 2 : Wiener homogène

```
liste <- simuler(sim_wiener, t7)
tracer_plusieurs(liste, paste("Wiener homogène :", n_traj, "trajectoires"))
```

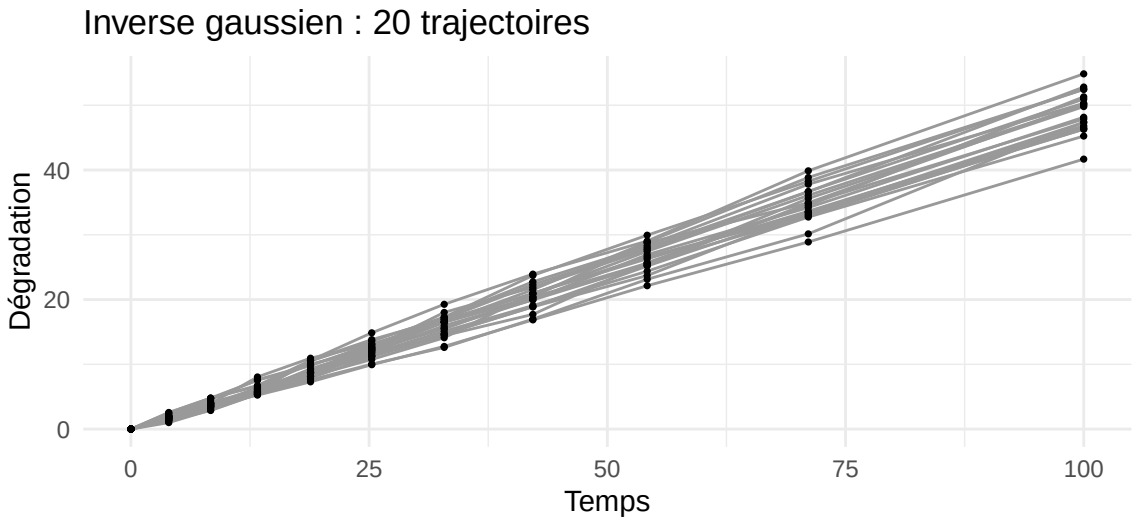



```
resultat(try(wiener_mle(liste), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.547 |
| sigma2 | 2.25 | 2.162 |
| s2 | 2.25 | 2.173 |

### II.3.3 Modèle 3 : inverse gaussien

```
liste <- simuler(sim_ig, t7)
tracer_plusieurs(liste, paste("Inverse gaussien :", n_traj, "trajectoires"))
```

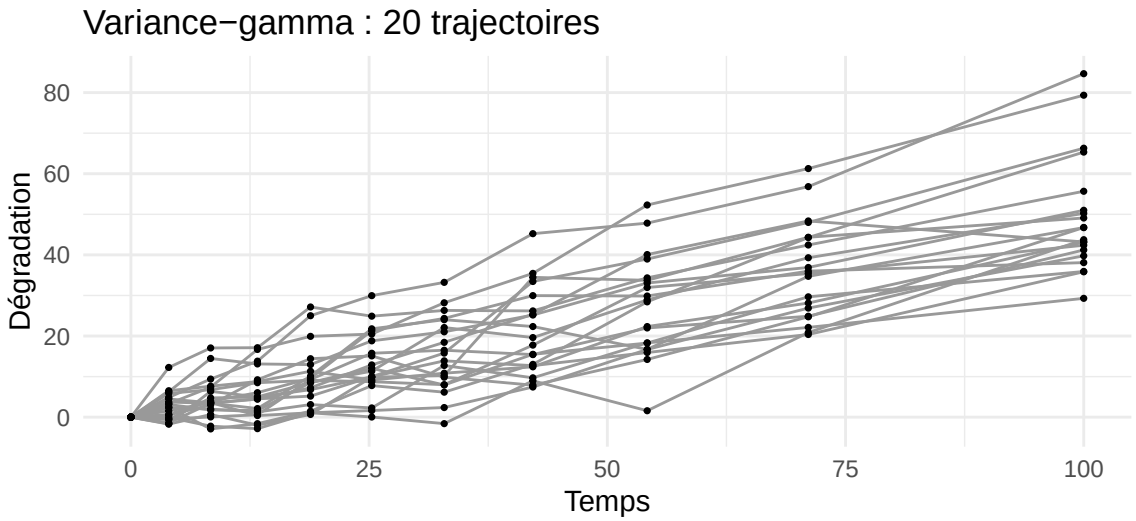


```r
resultat(try(ig_mle(liste), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.489 |
| eta | 2.0 | 1.770 |
| eta_unbiased | 2.0 | 1.743 |

### II.3.4 Modèle 4 : variance-gamma

```r
liste <- simuler(sim_vg, t7)
tracer_plusieurs(liste, paste("Variance-gamma :", n_traj, "trajectoires"))
```

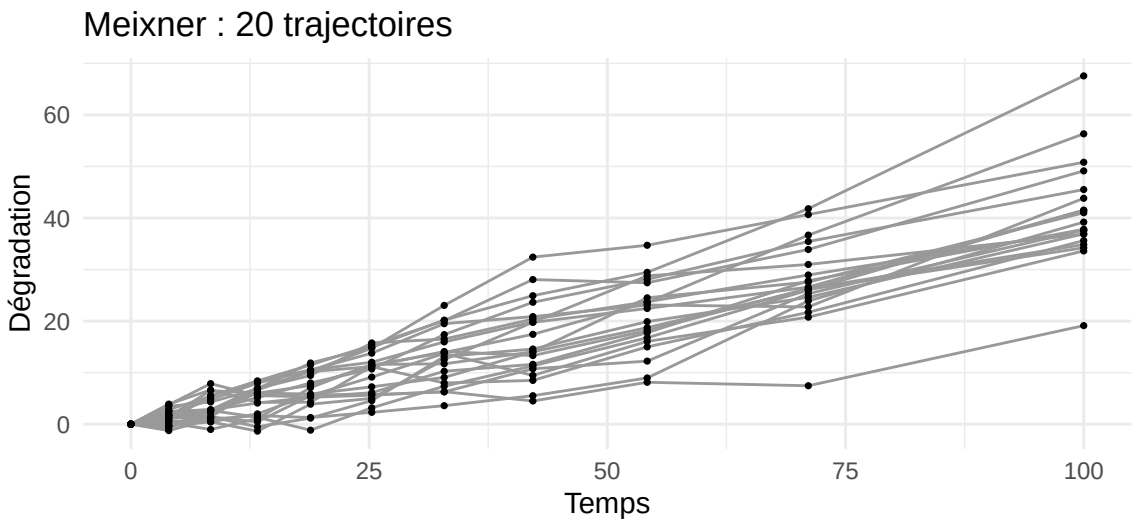



```r
resultat(try(vg_mle(liste), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.494 |
| sigma | 1.5 | 1.506 |
| nu | 2.0 | 1.867 |

### II.3.5 Modèle 5 : Meixner

```r
liste <- simuler(sim_meixner, t7)
tracer_plusieurs(liste, paste("Meixner :", n_traj, "trajectoires"))
```

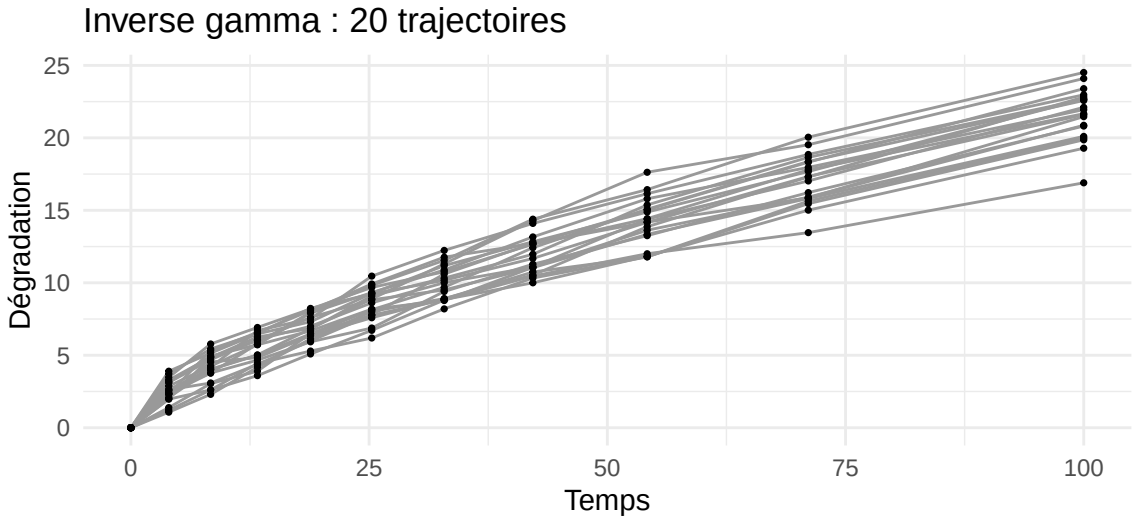



```
resultat(try(meixner_mle(liste), silent = TRUE), vrai_meixner)
```

**Estimation impossible** : no interior maximum: all solutions are at the boundary (shifted gamma or Gaussian limit)

### II.3.6 Modèle 6 : inverse gamma

```
liste <- simuler(sim_igam, t7)
tracer_plusieurs(liste, paste("Inverse gamma :", n_traj, "trajectoires"))
```



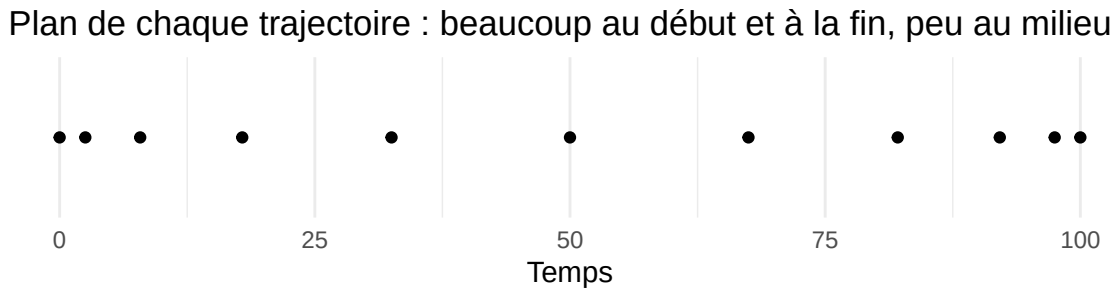

```
resultat(try(igam_mle(liste), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 0.878 |
| alpha | 1.0 | 0.831 |
| beta | 1.5 | 1.454 |

## II.4 Cas 8 : inspections mixtes

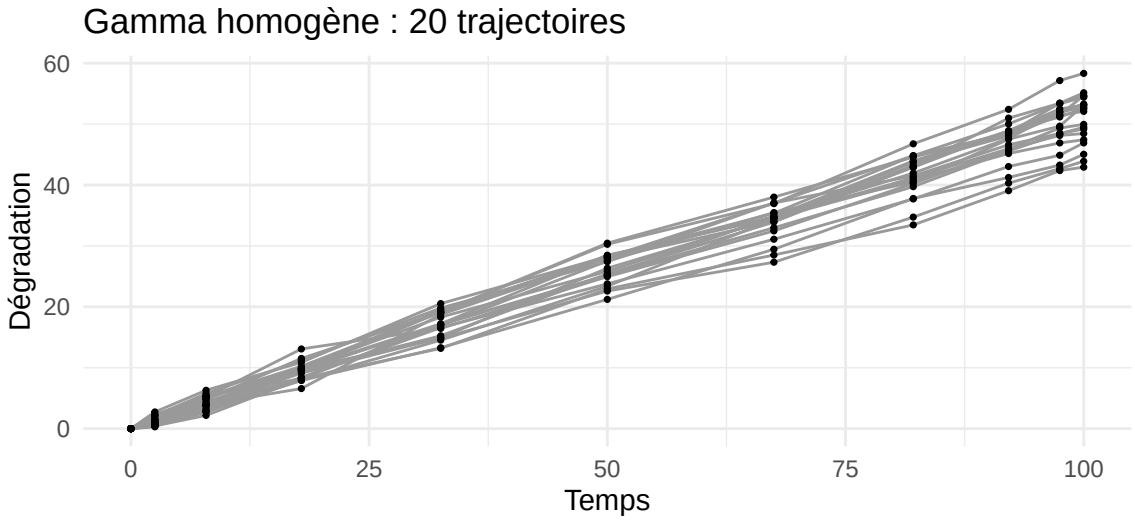
20 trajectoires indépendantes, chacune inspectée 10 fois jusqu'à 100. Chaque trajectoire est un `data.frame(t, X)`, et les fonctions du package reçoivent la liste de ces tableaux.

```
set.seed(8)
t8 <- inspection_times("dec-inc", n = 10, tmax = tmax)$time
tracer_plan(t8, "Plan de chaque trajectoire : beaucoup au début et à la fin, peu au milieu")
```




### II.4.1 Modèle 1 : gamma homogène

```
liste <- simuler(sim_gamma, t8)
tracer_plusieurs(liste, paste("Gamma homogène :", n_traj, "trajectoires"))
```

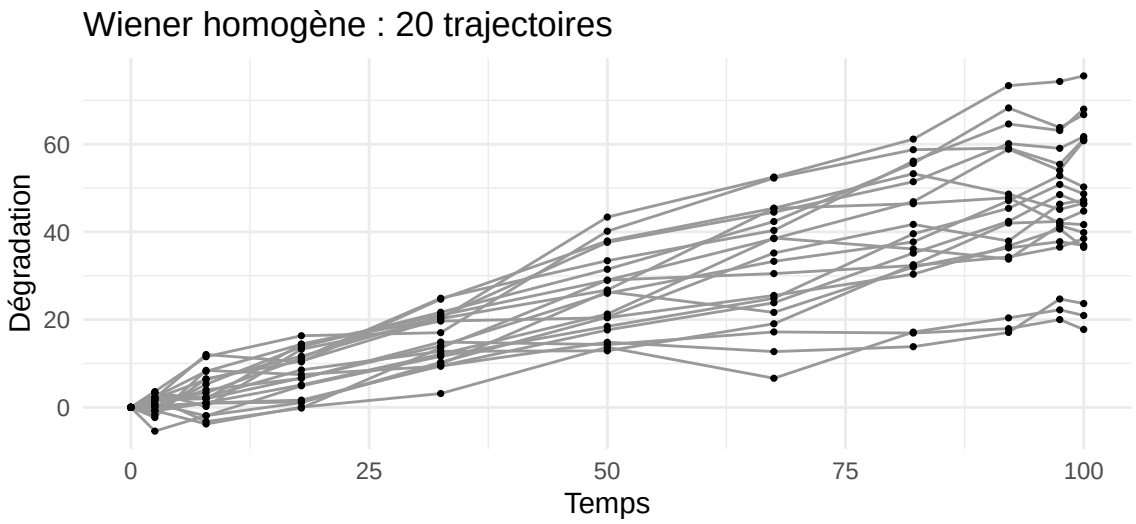



```
resultat(try(gamma_mle(liste), silent = TRUE), vrai_gamma)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.5 | 1.394 |
| b | 3.0 | 2.741 |

### II.4.2 Modèle 2 : Wiener homogène

```
liste <- simuler(sim_wiener, t8)
tracer_plusieurs(liste, paste("Wiener homogène :", n_traj, "trajectoires"))
```

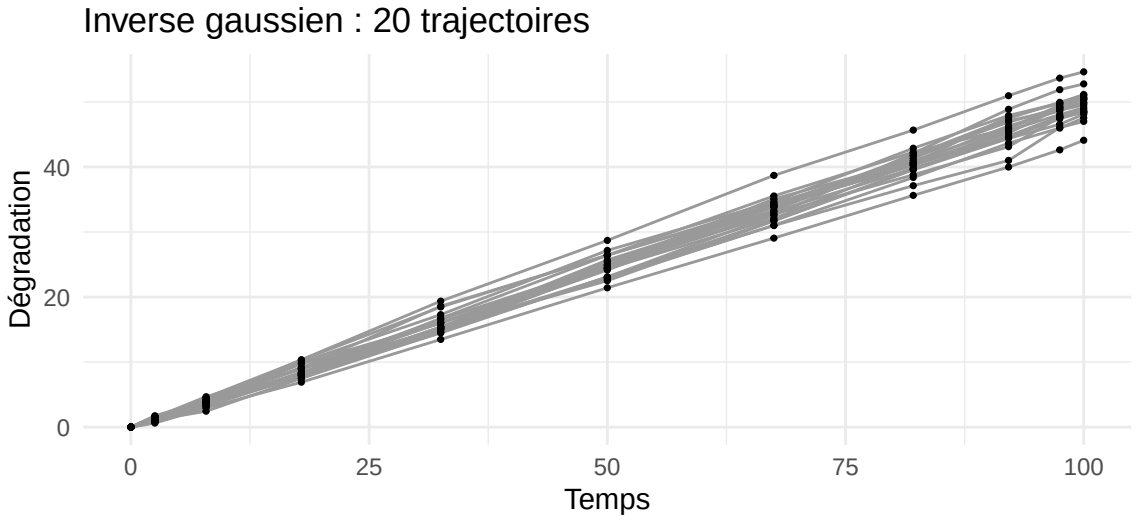



```
resultat(try(wiener_mle(liste), silent = TRUE), vrai_wiener)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.50 | 0.467 |
| sigma2 | 2.25 | 2.467 |
| s2 | 2.25 | 2.480 |

### II.4.3 Modèle 3 : inverse gaussien

```
liste <- simuler(sim_ig, t8)
tracer_plusieurs(liste, paste("Inverse gaussien :", n_traj, "trajectoires"))
```

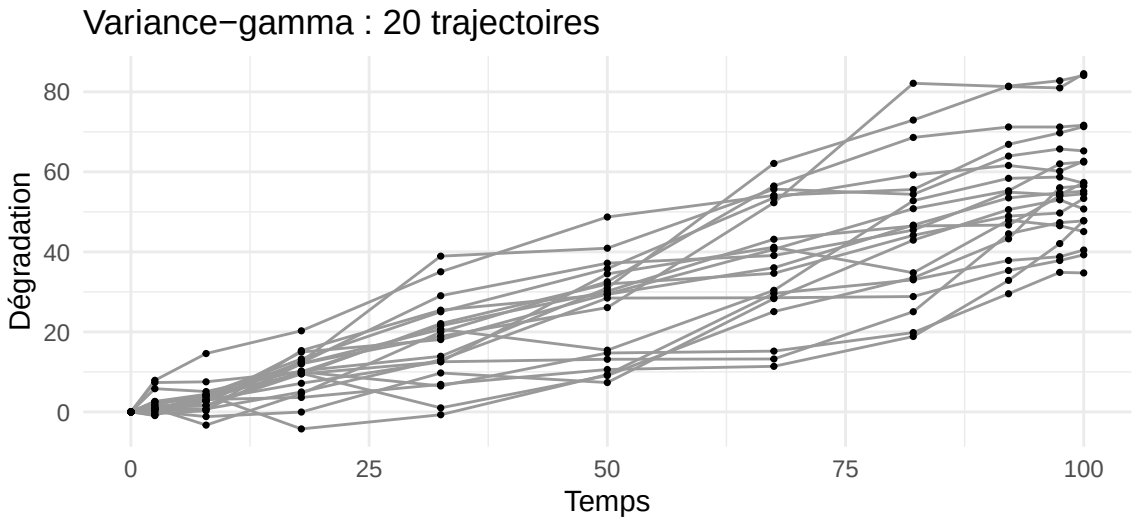


```r
resultat(try(ig_mle(liste), silent = TRUE), vrai_ig)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| mu | 0.5 | 0.495 |
| eta | 2.0 | 1.966 |
| eta_unbiased | 2.0 | 1.937 |

### II.4.4 Modèle 4 : variance-gamma

```r
liste <- simuler(sim_vg, t8)
tracer_plusieurs(liste, paste("Variance-gamma :", n_traj, "trajectoires"))
```

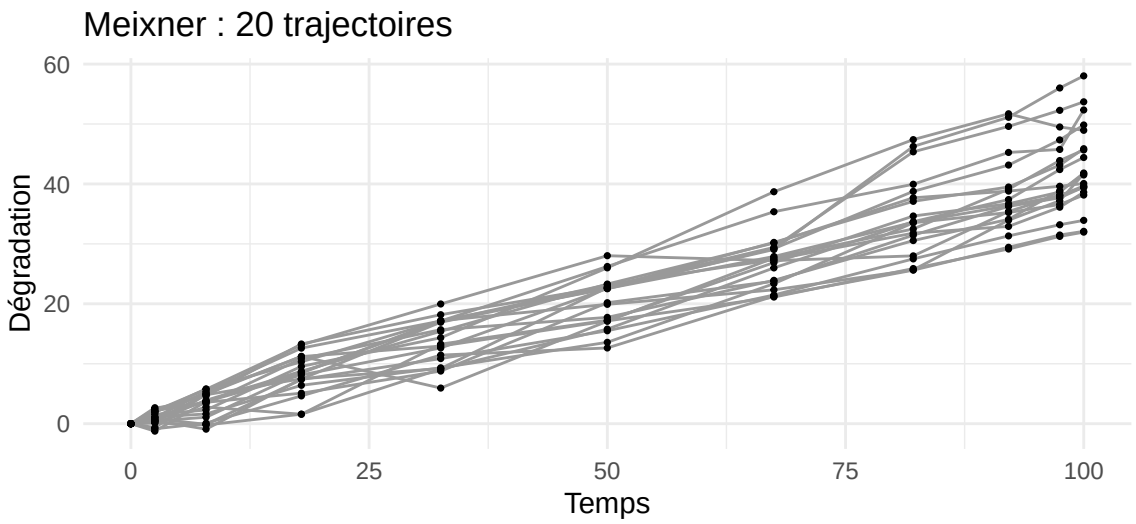



```r
resultat(try(vg_mle(liste), silent = TRUE), vrai_vg)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 0.5 | 0.571 |
| sigma | 1.5 | 1.502 |
| nu | 2.0 | 2.418 |

### II.4.5 Modèle 5 : Meixner

```r
liste <- simuler(sim_meixner, t8)
tracer_plusieurs(liste, paste("Meixner :", n_traj, "trajectoires"))
```

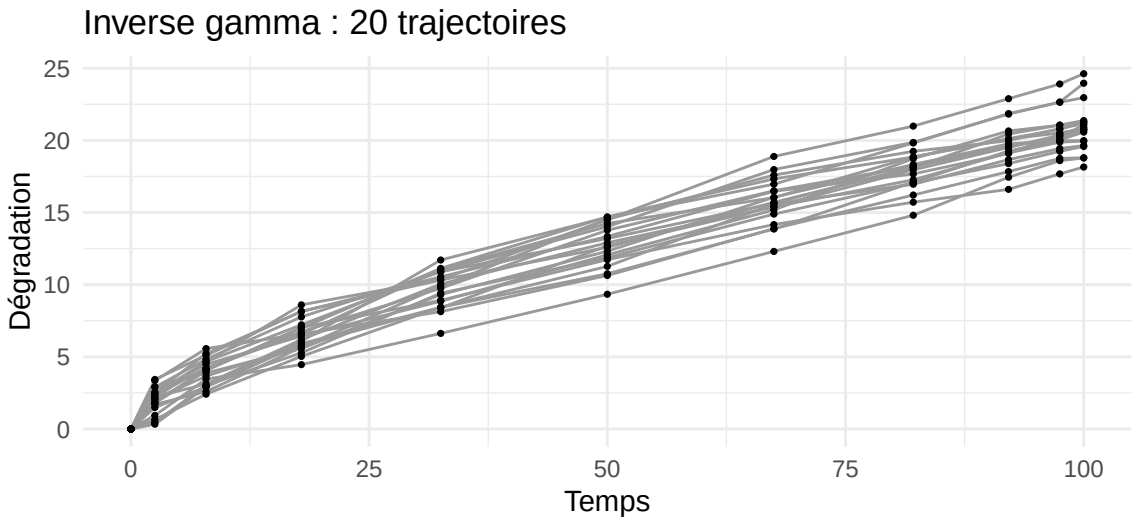



```
resultat(try(meixner_mle(liste), silent = TRUE), vrai_meixner)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| a | 1.0 | 2.589 |
| b | 0.3 | 0.034 |
| d | 1.5 | 0.215 |
| m | 0.2 | 0.419 |

#### II.4.6 Modèle 6 : inverse gamma

```
liste <- simuler(sim_igam, t8)
tracer_plusieurs(liste, paste("Inverse gamma :", n_traj, "trajectoires"))
```



```
resultat(try(igam_mle(liste), silent = TRUE), vrai_igam)
```

| Paramètre | Vraie | Estimée |
|---|---|---|
| theta | 1.0 | 1.156 |
| alpha | 1.0 | 0.986 |
| beta | 1.5 | 1.463 |